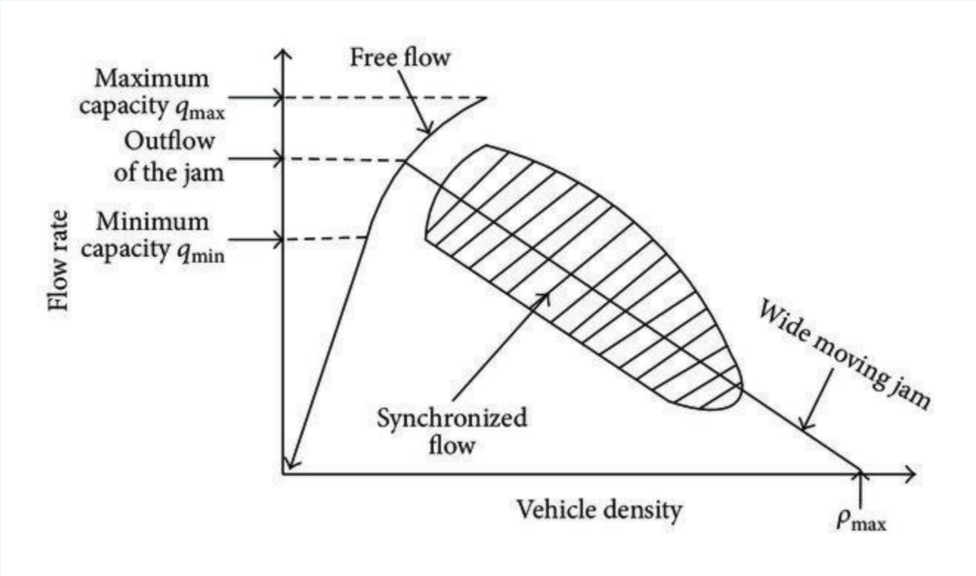
Modeling and Predicting Behaviors of Inductive Traffic Loops

Cooper Amador-Risch, **Tushar Mehta,** Swetha Menon

Boris Kerner's Traffic Theory

Free Flow → Synchronized Flow → Wide Moving Jam



Inductive Traffic Loops

Image from: https://www.gcelab.com/blog/what-is-inductive-loop-detector-and-advantages

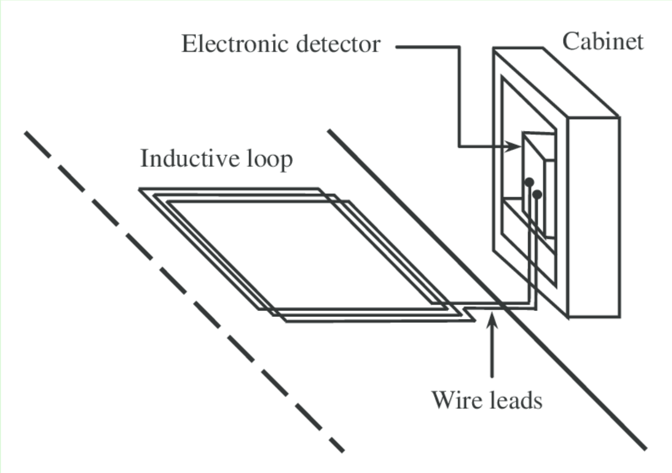


Image from: https://www.gcelab.com/blog/what-is-inductive-loop-detector-and-advantages

Introduction and Theory

Electromagnetic Theory

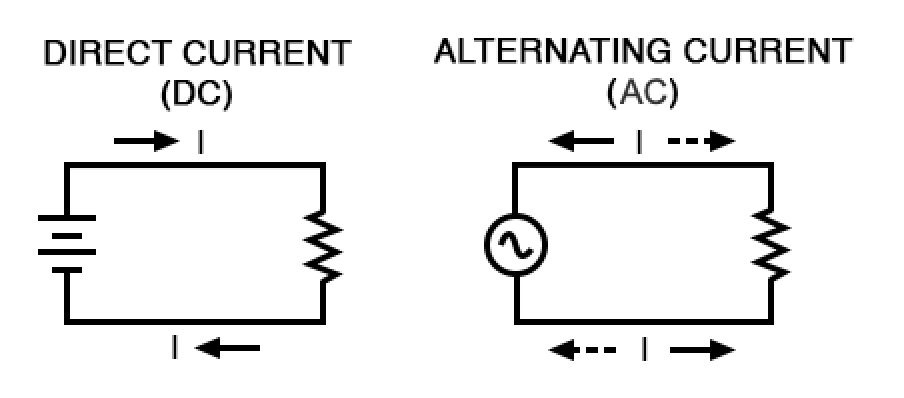


Image from: https://www.allaboutcircuits.com/textbook/alternating-current/chpt-1/what-is-alternating-current-ac/

Maxwell's Equations - EM Theory

Ampère’s Law: $$\oint B \cdot dl = \mu_0 I_n$$

Electric currents generate magnetic fields

Gauss’s Law for Electricity: $$\oint E \cdot dA = \frac{Q}{\epsilon_0}$$

Electric Fields form electric charges

Gauss’s Law for Magnetism: $$\oint B \cdot dA = 0$$

No magnetic monopoles

Induction

- Total electric field that passes through a surface
- Flux Equation $\phi_B = BAcos \cdot (\theta)$
- Changing magnetic field induces an EMF force in a conductor

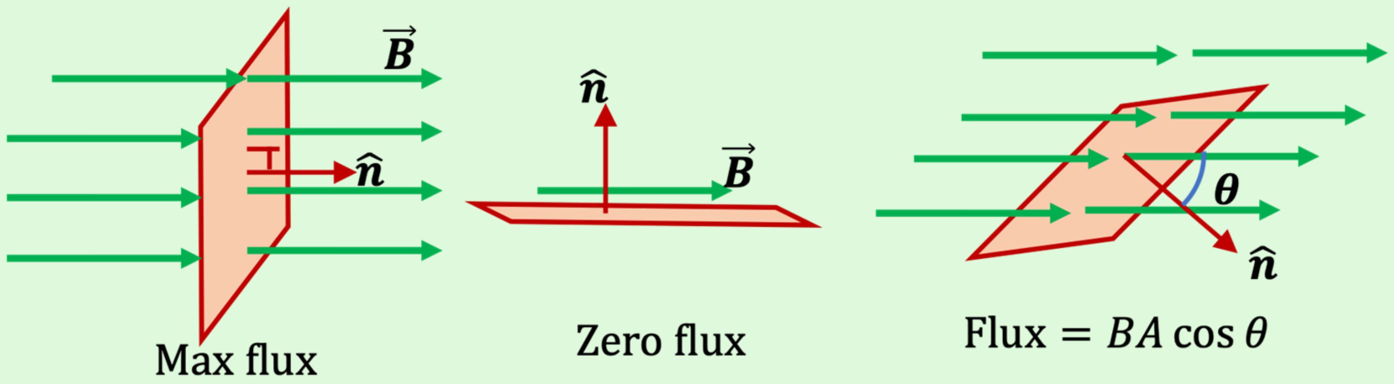


Image: https://www.physicsbootcamp.org/Magnetic-Flux.html

Faraday's Law

- Changing magnetic fields induce currents
- Lenz's Law
- Opposition ensures a resistance to sudden changes

$$\epsilon = -\frac{d\phi_B}{dt}$$

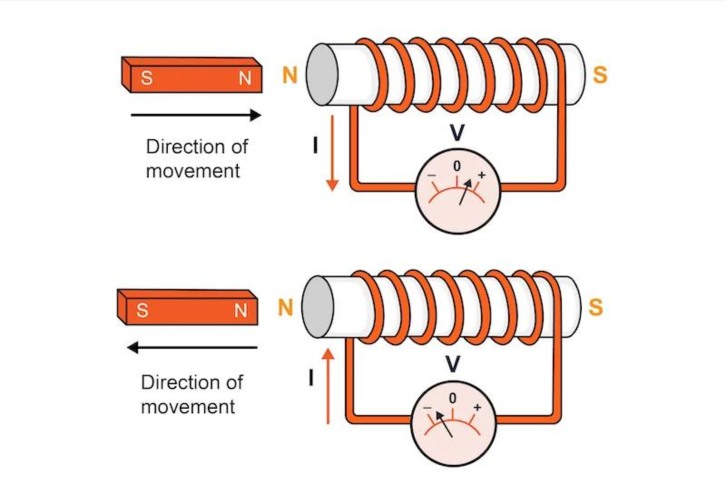


Image: https://www.allaboutcircuits.com/tools/lenz-law-calculator-faradays-law-calculator/

Self Inductance

- Specific case of Faraday's law

$$\epsilon_2 = -M\frac{dI_1}{dt}$$

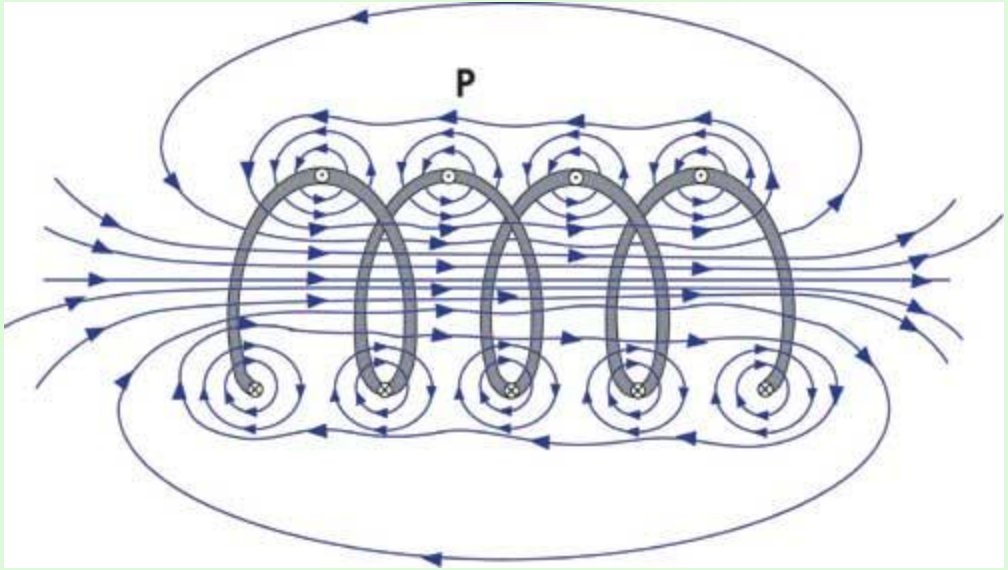


Image from: https://www.nde-ed.org/Physics/Electricity/selfinductance.xhtml

Ferromagnetism and Magnetic Moments

- Collective alignment of atomic magnetic moments
- Exchange interaction

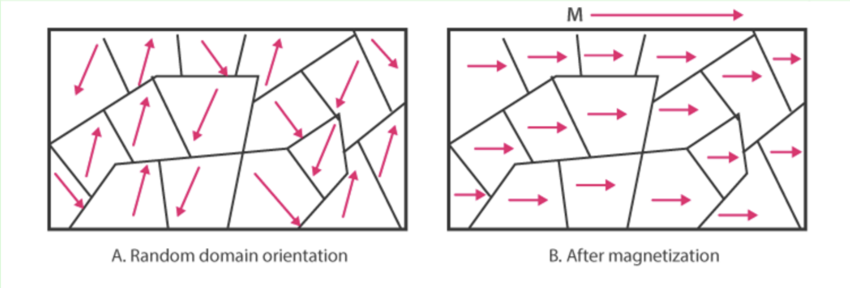


Image from: https://byjus.com/jee/ferromagnetic-materials/

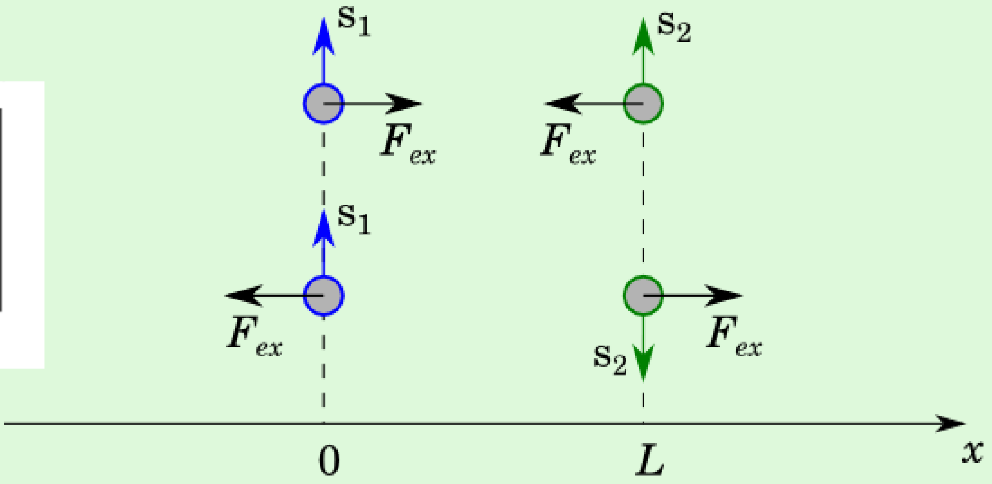


Image from: https://physics.stackexchange.com/questions/18395/simple-description-of-exchange-interaction

Loop Circuit

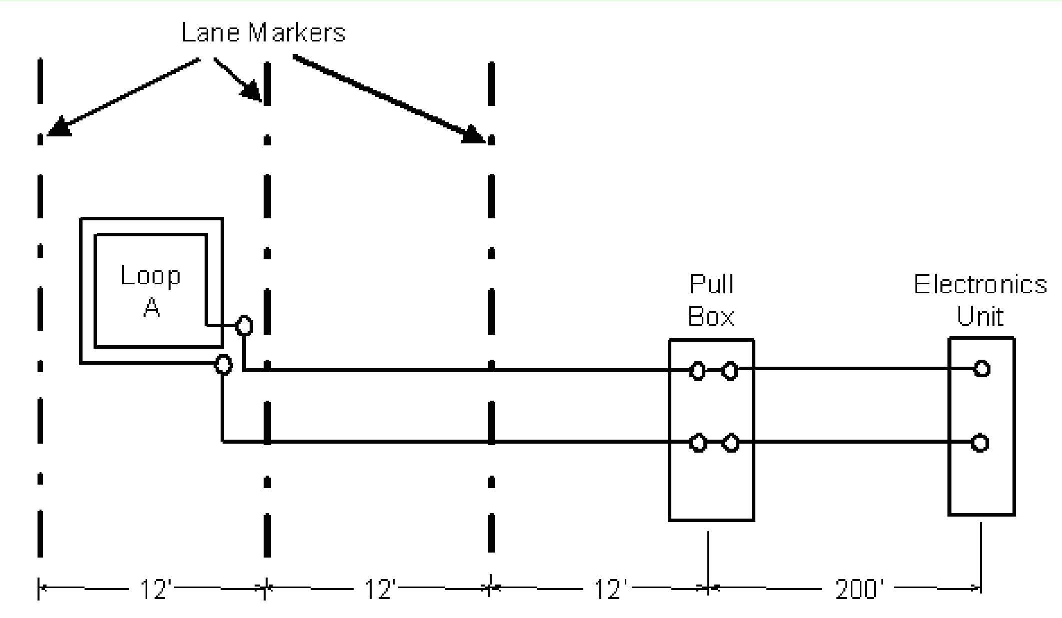
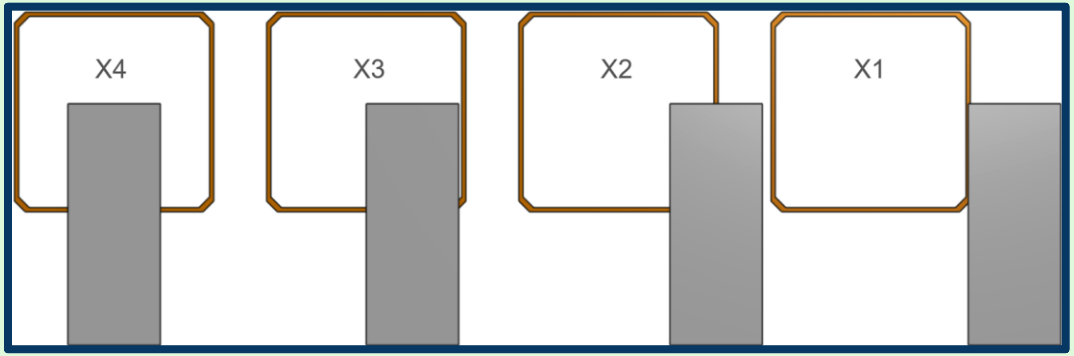


Image from:
https://www.fhwa.dot.gov/publications/research/operations/its/06108/02.cfm

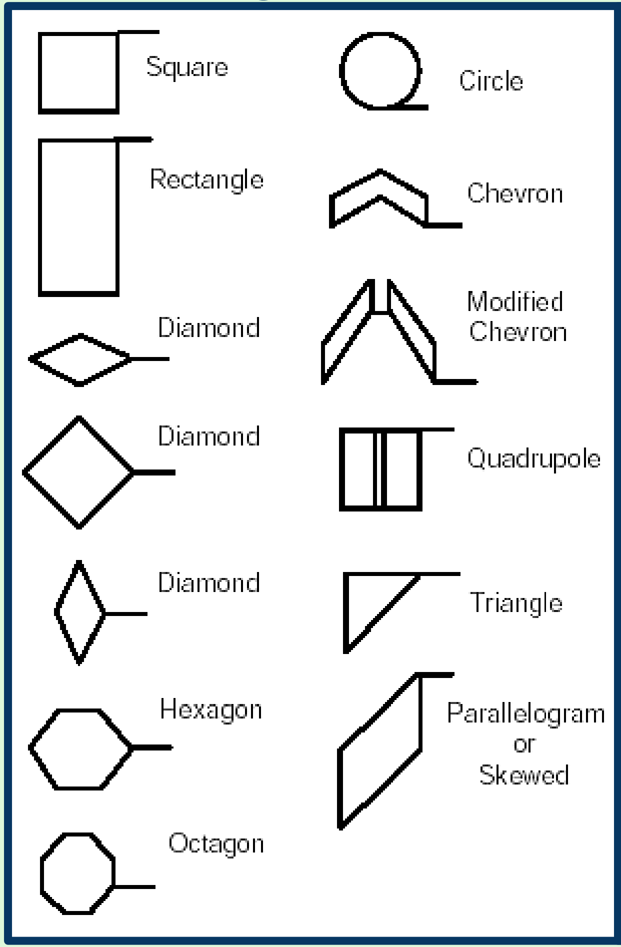
Goal

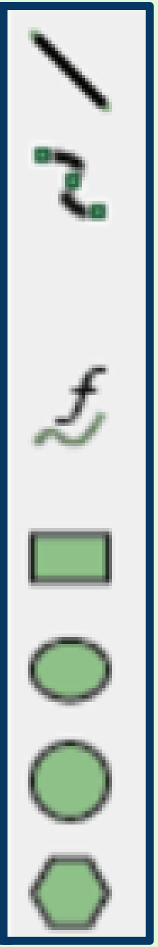
- Detect the Lane Positioning of a Motorcycle over the Inductive Traffic Loop

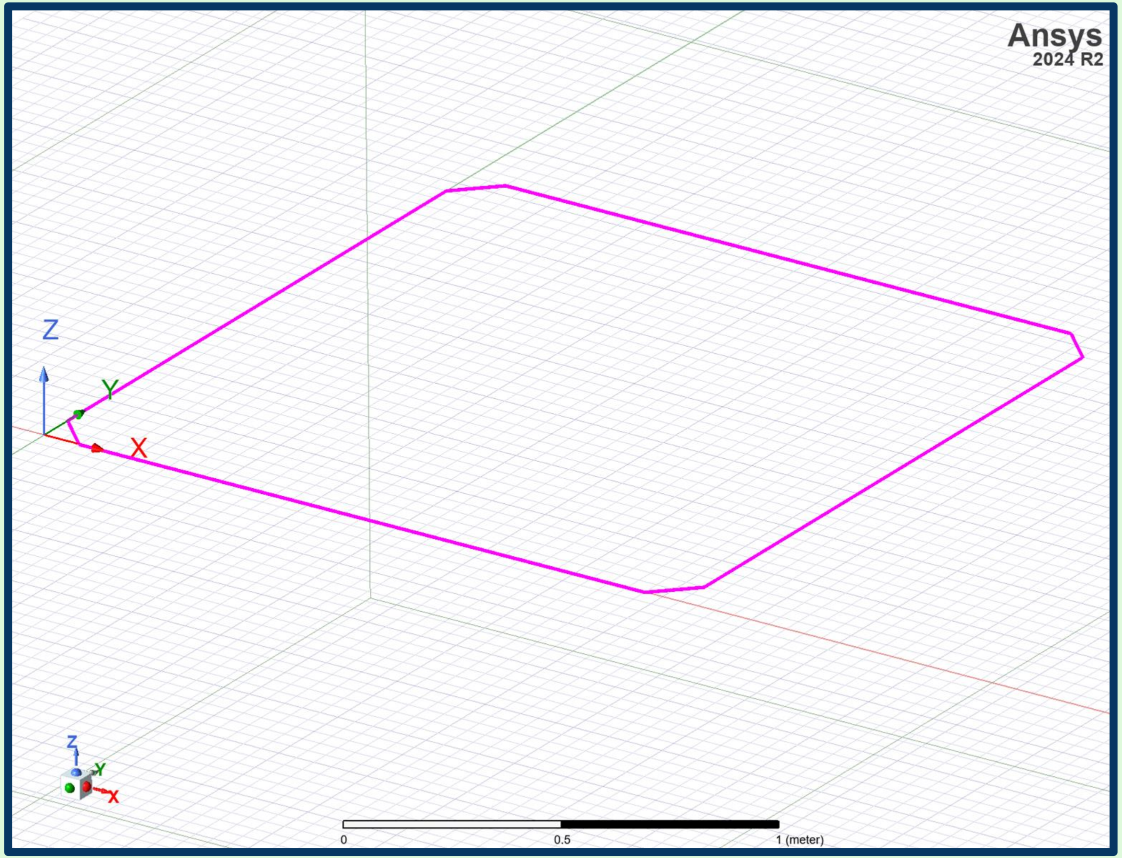


Simulation

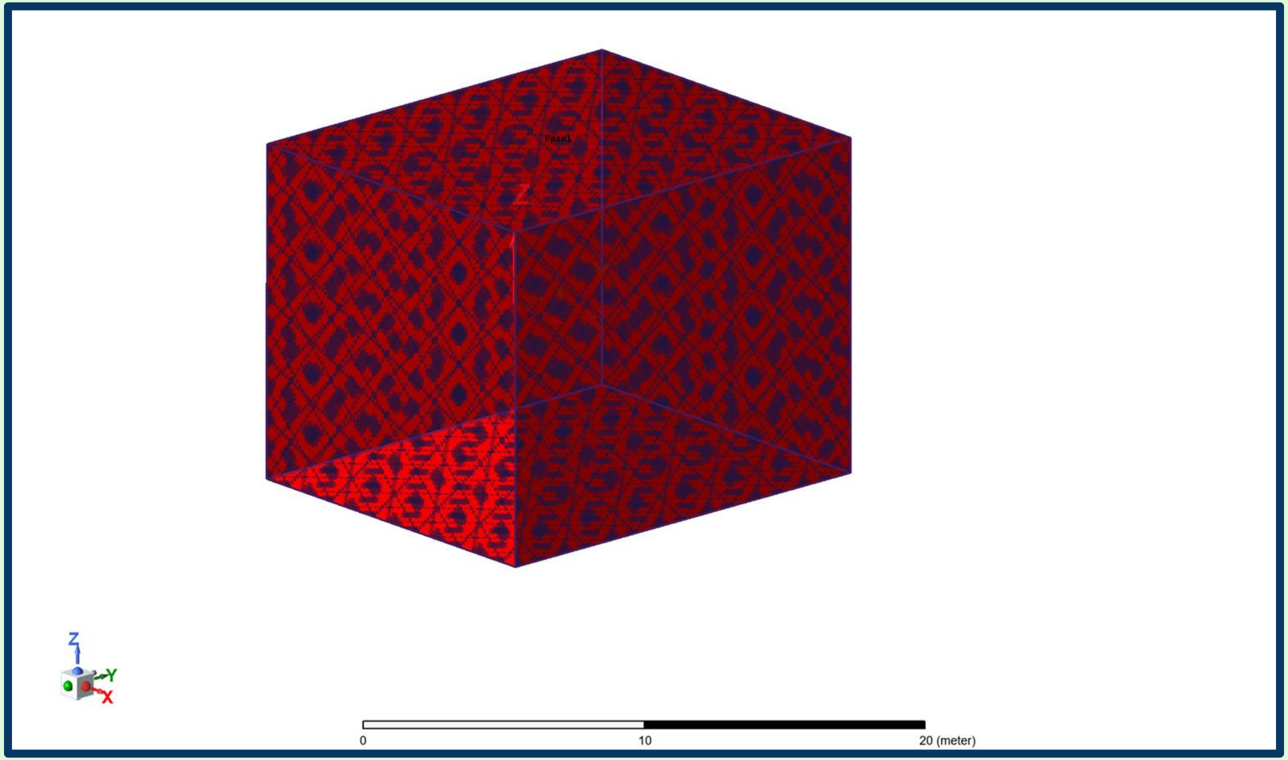
Modeling the Loop



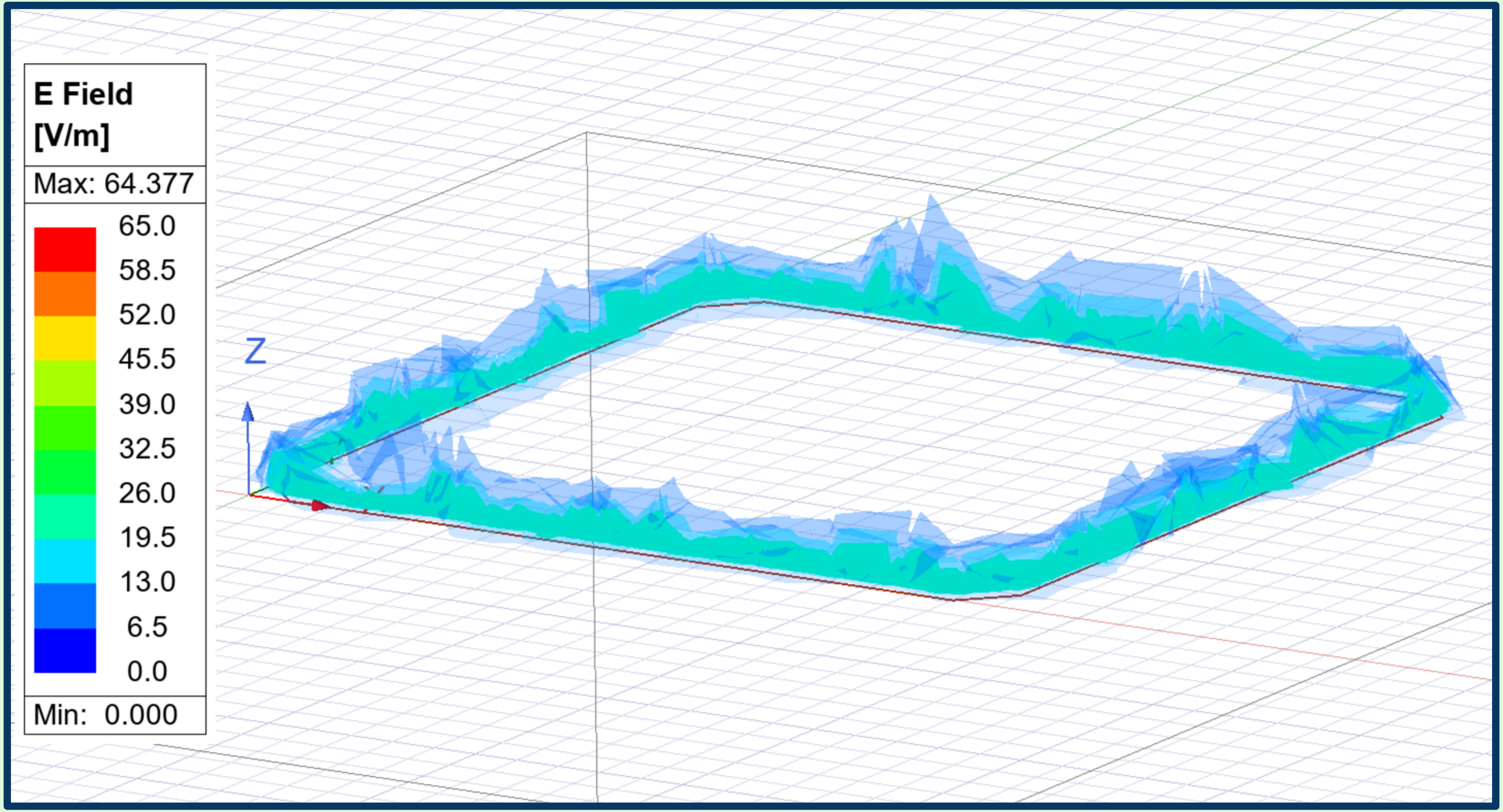




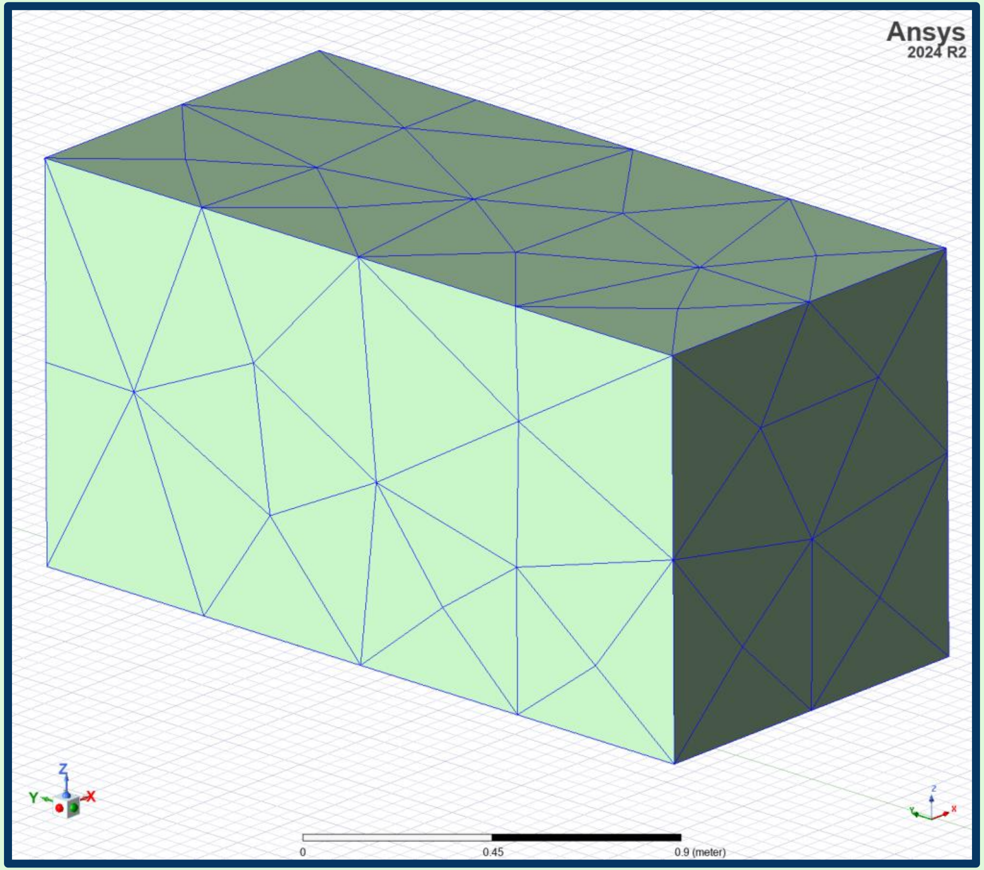
Boundary Conditions



Excitation


E Field
[V/m]
Max: 64.377
65.0
58.5
52.0
45.5
39.0
32.5
26.0
19.5
13.0
6.5
0.0
Min: 0.000
Z

Modeling the Vehicle



Solver Setup

-Frequency Sweep

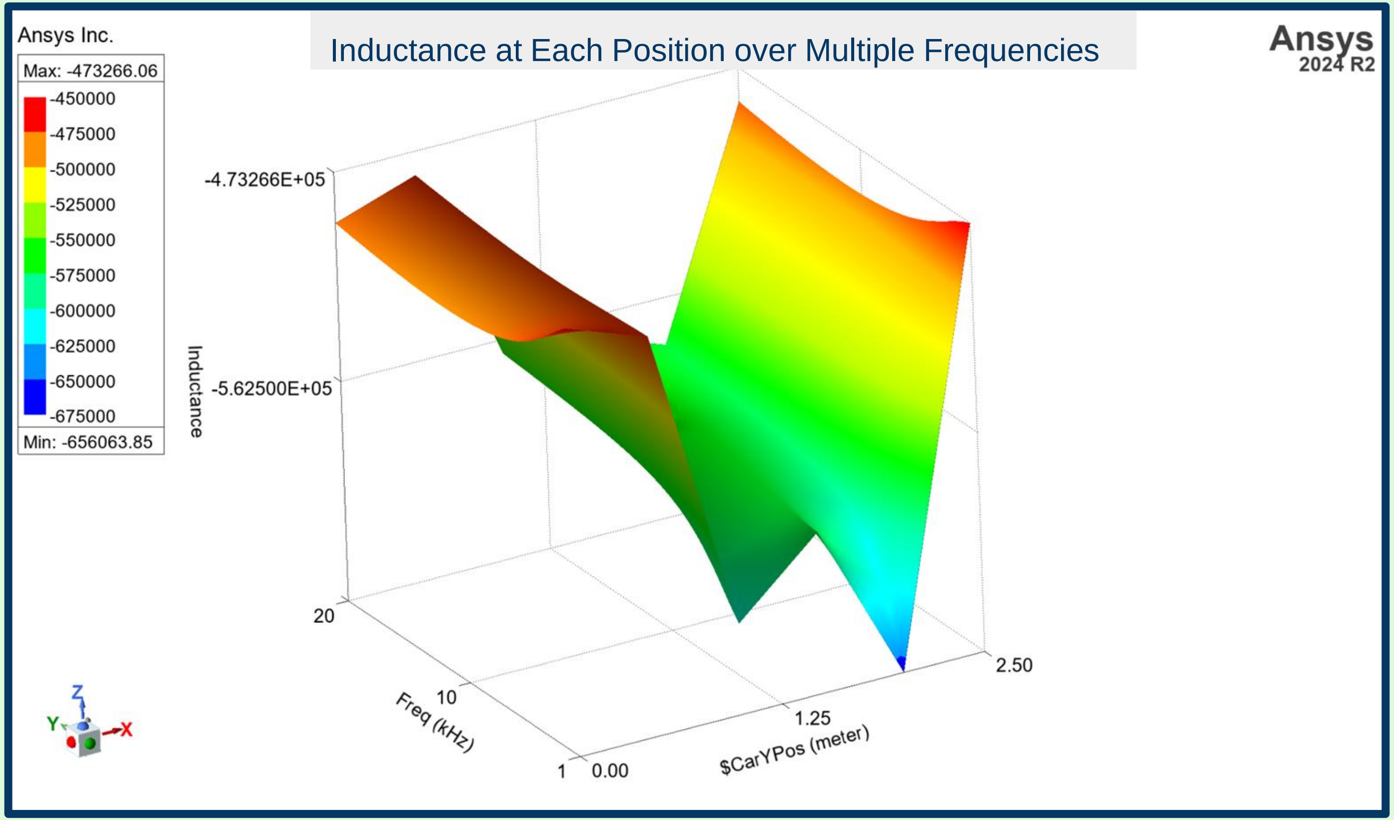
-Balancing Settings

-Parametric Sweep

-Finite Element Method (FEM)

Frequency Sweep

-1kHz to 20kHz



Balancing Settings

-High computational cost

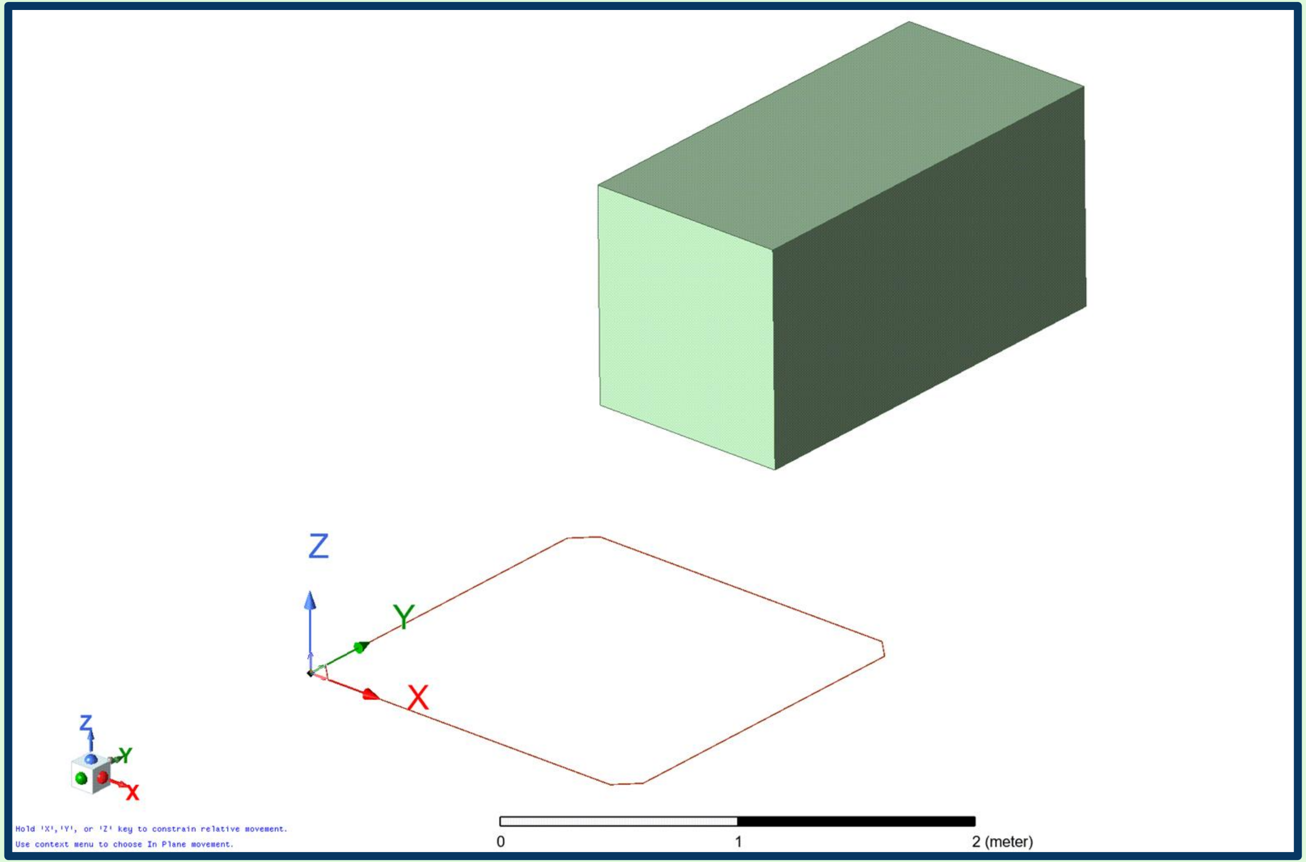
- Hours and days

- at most an hour

-accuracy and efficiency

Parametric Sweep

$$Inductance = \frac{im(Z(1,1))}{2 * \pi * freq}$$



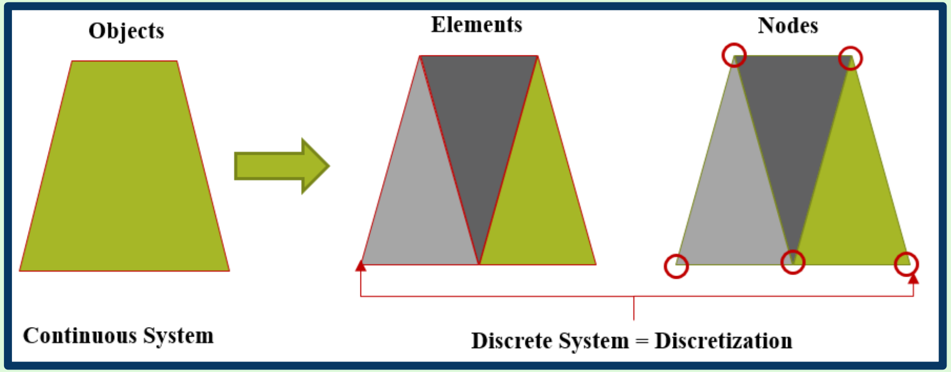
Finite Element Method (FEM)

-Breaks large problem into smaller ones

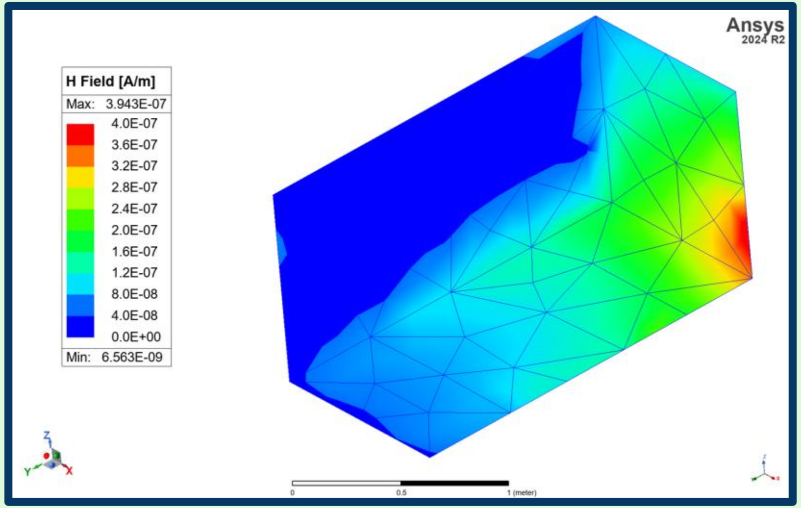
-Similar to perturbation problems

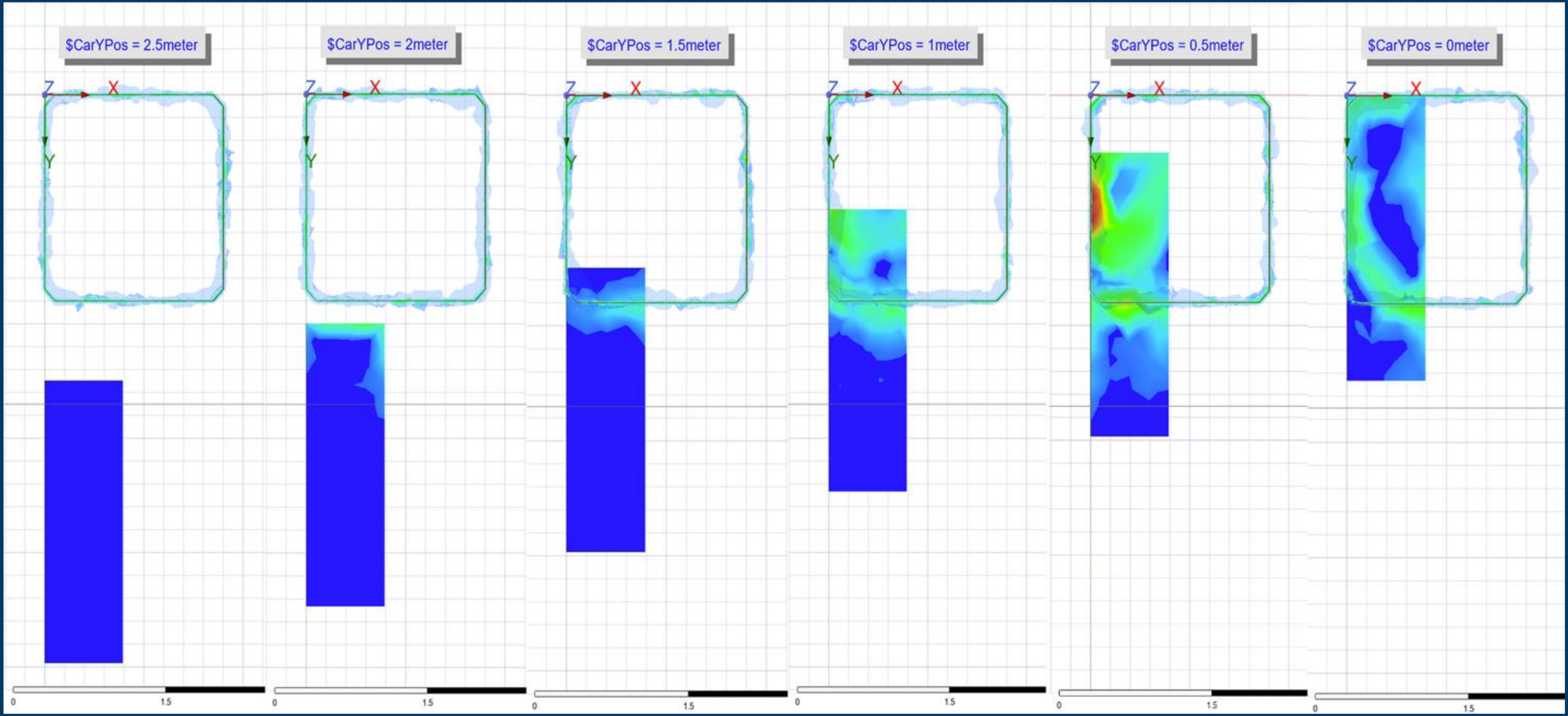
-Choosing right element size

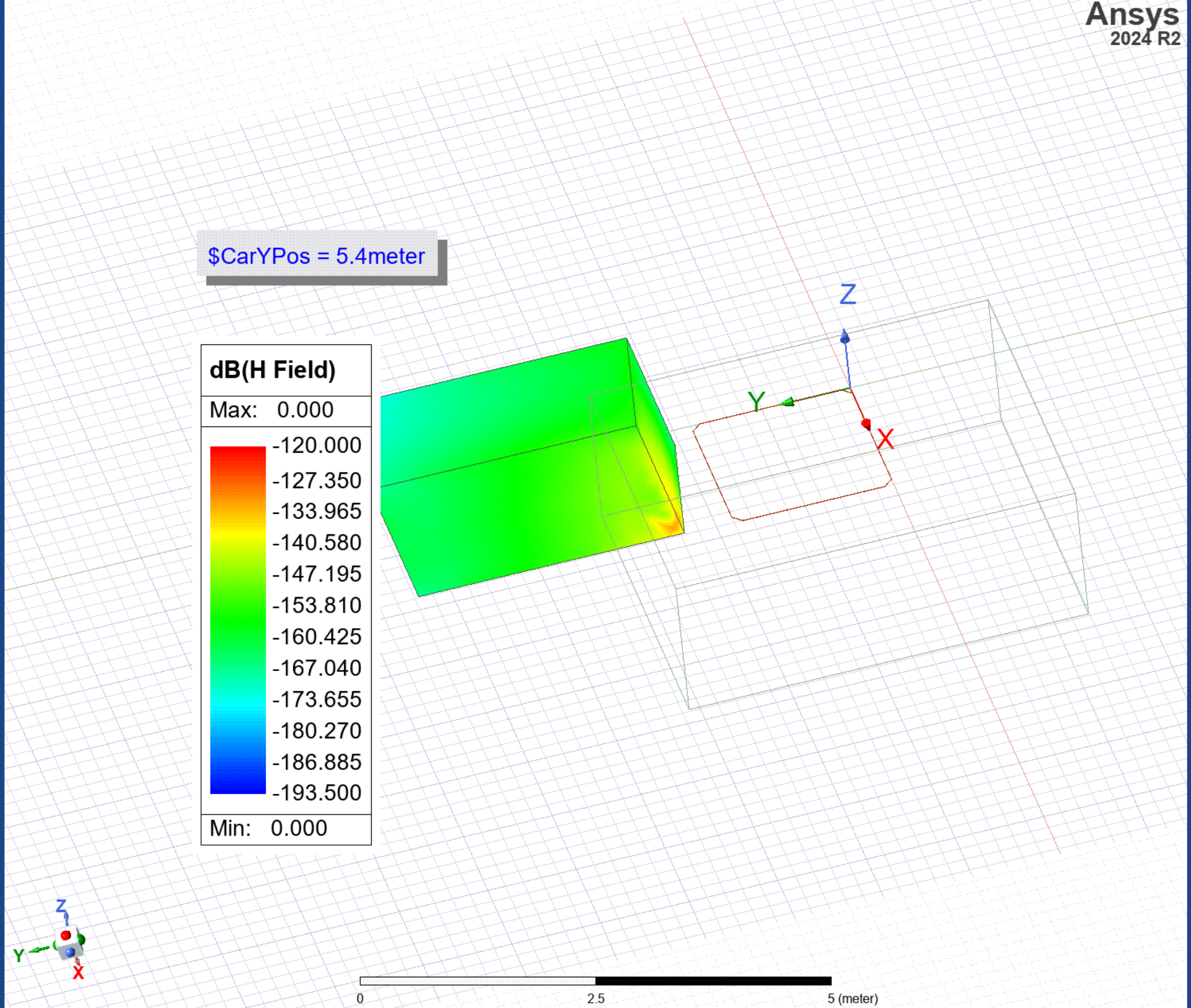
-Adaptive Meshing



https://www.graspengineering.com/hello-world-2/



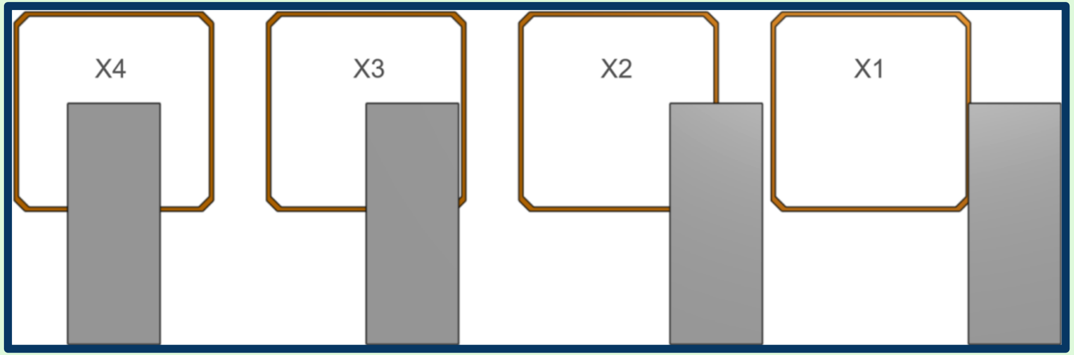

$CarYPos = 2.5meter
$CarYPos = 2meter
$CarYPos = 1.5meter
$CarYPos = 1meter
$CarYPos = 0.5meter
$CarYPos = 0meter


Ansys
2024 R2
$CarYPos = 5.4meter
dB(H Field)
Max: 0.000
-120.000
-127.350
-133.965
-140.580
-147.195
-153.810
-160.425
-167.040
-173.655
-180.270
-186.885
-193.500
Min: 0.000
Z
Y
X
0
2.5
5 (meter)

Data Generation

Original Data Set

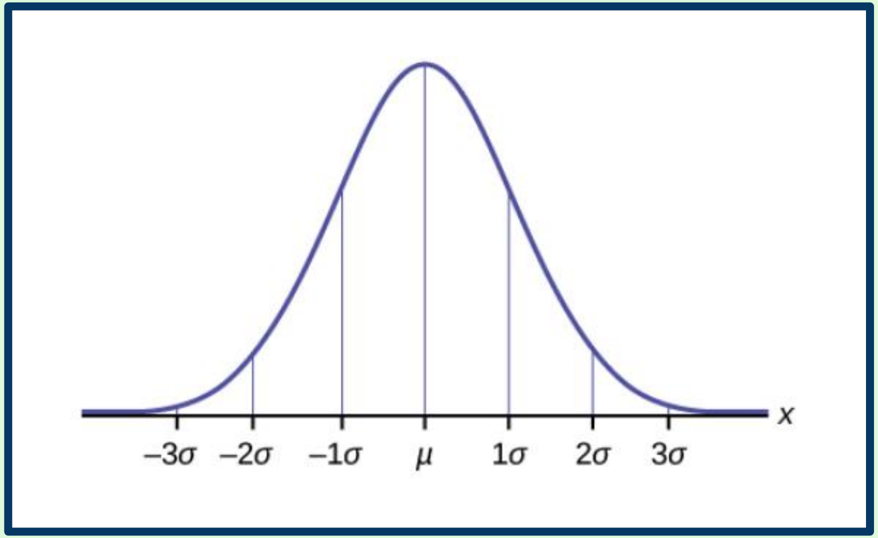
x width	0.84 meters
y width	2.27 meters
z width	1.085 meters



	$CarYPos [meter]	mean(Inductance) Setup2 : LastAdaptive
1	0.000000	-582237.002705
2	0.500000	-601920.418868
3	1.000000	-583920.344462
4	1.500000	-487178.241617
5	2.000000	-573738.132871
6	2.500000	-578156.496581

Hor. Position	ΔInductance
x1	77268
x2	93986
x3	103298
x4	114742

First Test: Gaussian Noise Distribution

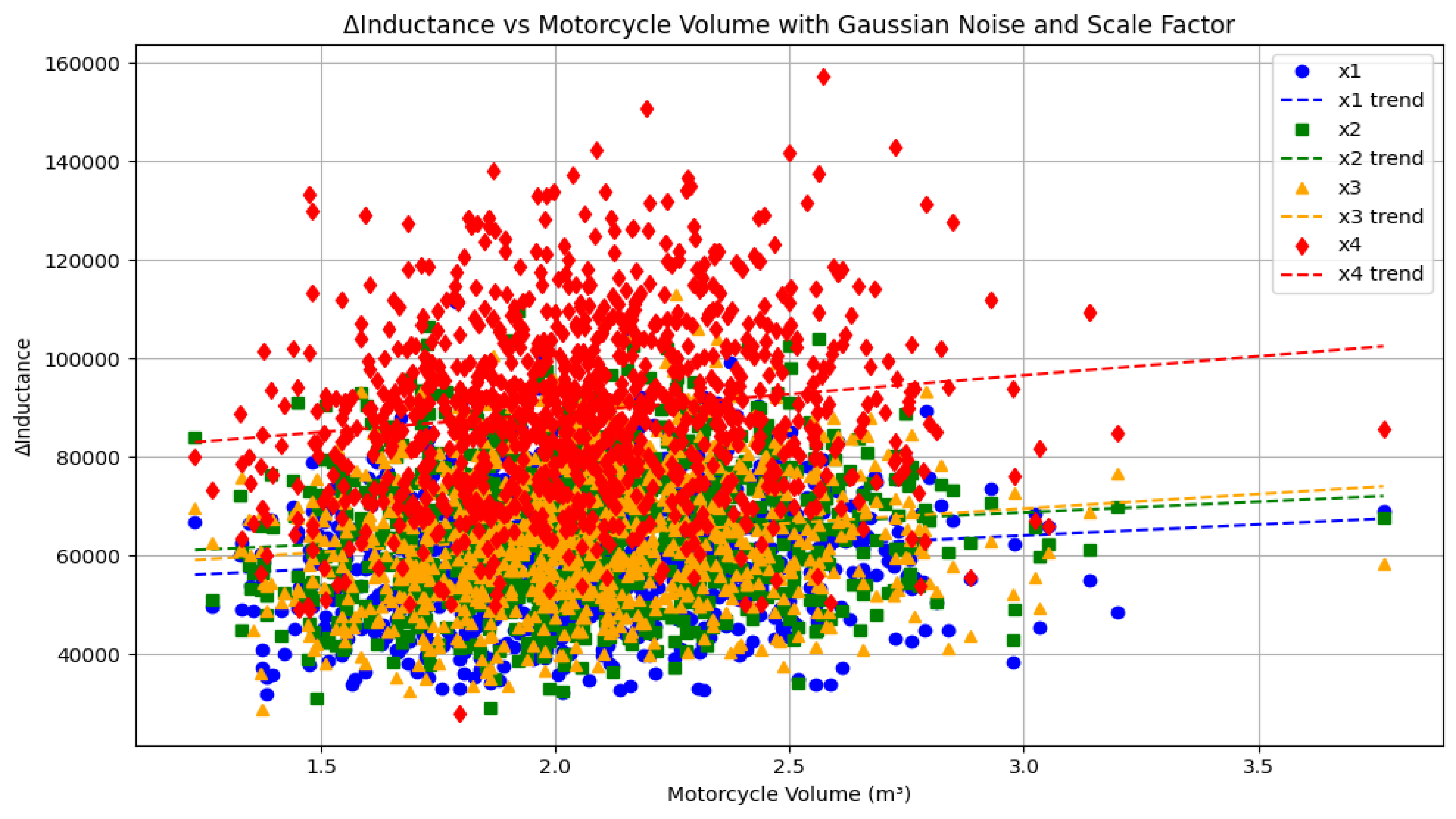


Barbara Illowsky, Susan Dean. *Introductory Statistics.* Openstax, Rice University. Houston, Texas. 2018.

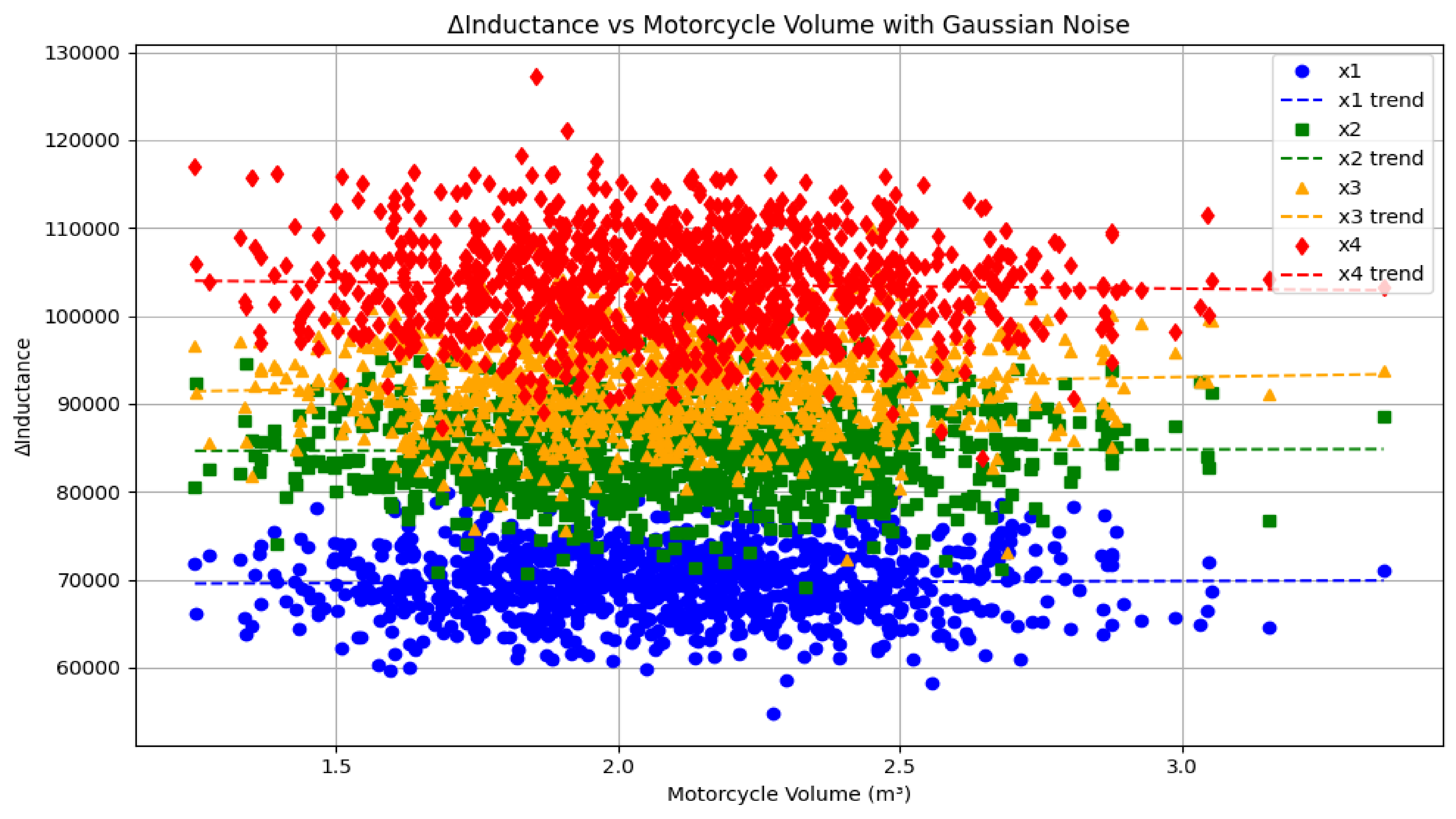
$$\text{scale factor} = \frac{\text{volume}}{\text{base volume}}$$

$$f(x) = \frac{1}{\sigma * \sqrt{2 * \pi}} * e^{-\frac{1}{2} * \left(\frac{x - \mu}{\sigma}\right)^2}$$

$$\text{new inductance} = \text{base inductance} * \text{Gaussian noise} * \text{scale factor}$$

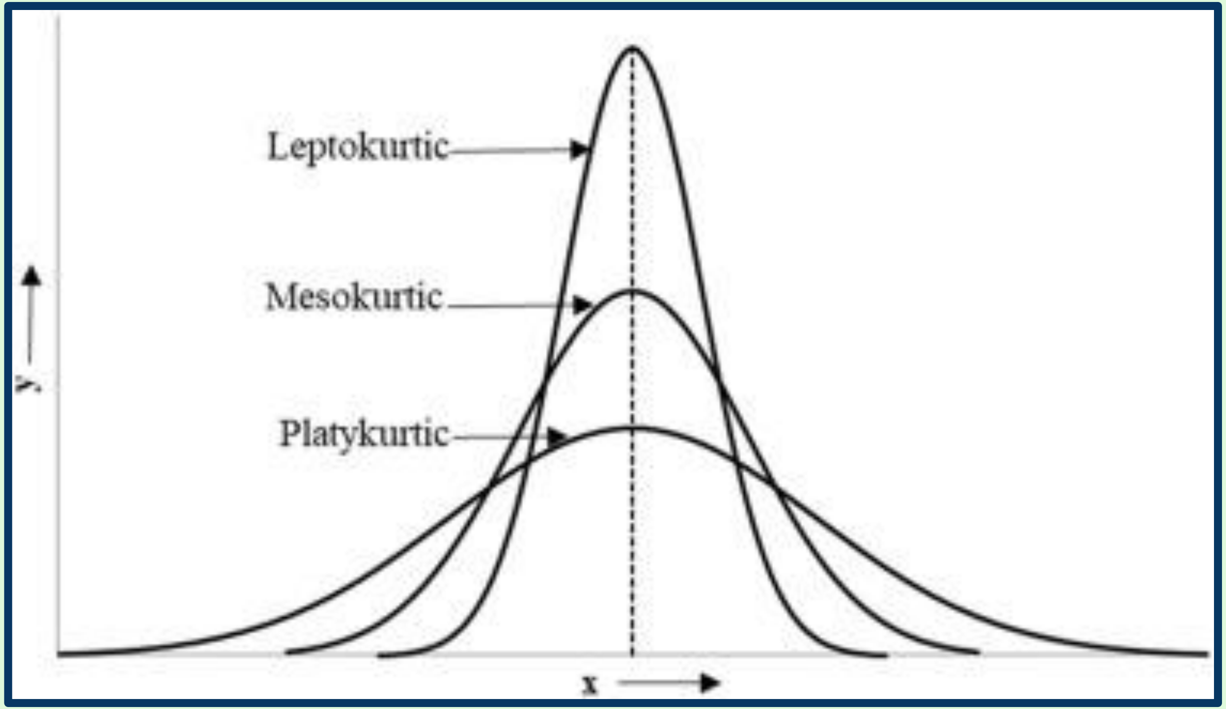


ΔInductance vs Motorcycle Volume with Gaussian Noise and Scale Factor
160000
140000
120000
100000
80000
60000
40000
ΔInductance
1.5
2.0
2.5
3.0
3.5
Motorcycle Volume (m³)
x1
x1 trend
x2
x2 trend
x3
x3 trend
x4
x4 trend


ΔInductance vs Motorcycle Volume with Gaussian Noise
130000
120000
110000
100000
90000
80000
70000
60000
ΔInductance
1.5
2.0
2.5
3.0
Motorcycle Volume (m³)
x1
x1 trend
x2
x2 trend
x3
x3 trend
x4
x4 trend

Second Test: Hyperbolic Secant Distribution

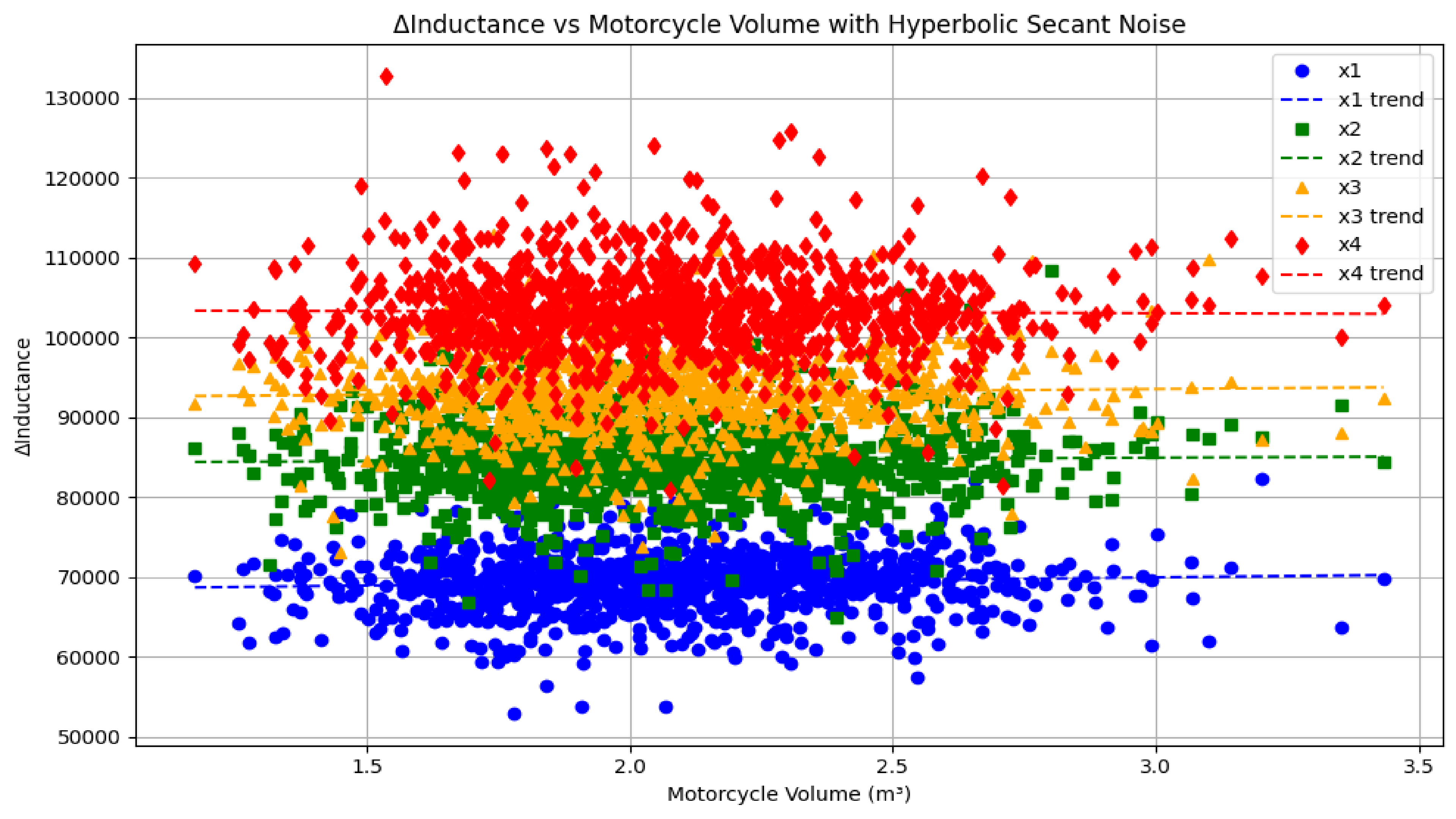
https://www.sciencedirect.com/book/9780128168189/highway-safety-analytics-and-modeling



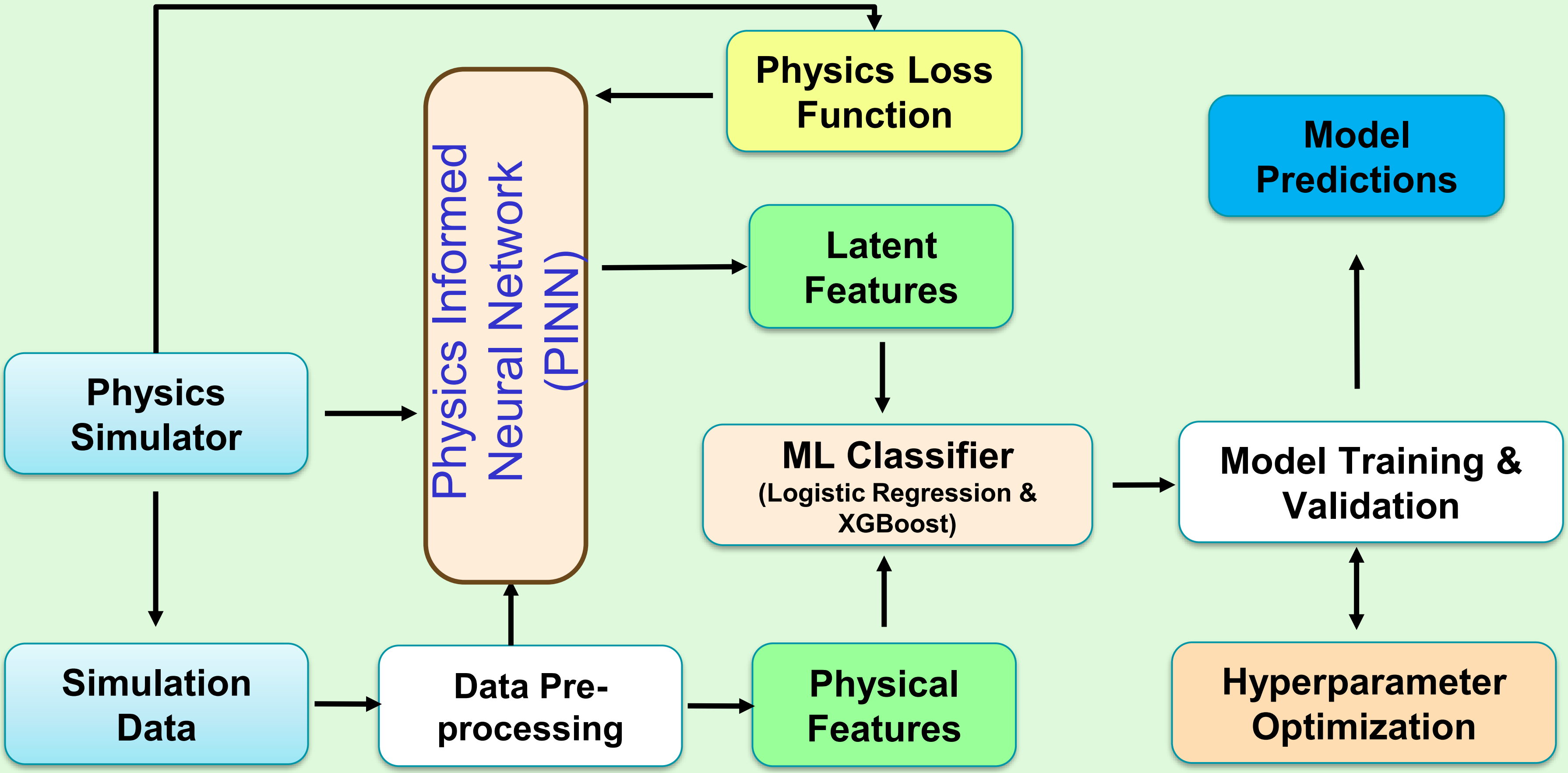
$$f(x) = \frac{1}{2}\text{sech}(\frac{\pi * x}{2})$$

$$F(x) = \frac{2}{\pi}\arctan(e^{\frac{\pi * x}{2}})$$

$$X = \frac{2}{\pi}\ln(\tan(\frac{\pi * U}{2})), \quad U \in (0,1)$$


ΔInductance vs Motorcycle Volume with Hyperbolic Secant Noise
ΔInductance
Motorcycle Volume (m³)
130000
120000
110000
100000
90000
80000
70000
60000
50000
1.5
2.0
2.5
3.0
3.5
x1
x1 trend
x2
x2 trend
x3
x3 trend
x4
x4 trend

Physics Informed ML Pipeline



Data Pre-Processing

Label Encoding

```python
# 1) Load & melt  x1 x2 x3 x4
df = pd.read_csv('motorcycle_data_cross_section.csv')
melted = df.melt(value_vars=['x1', 'x2', 'x3','x4'], var_name='position', value_name='inductance')

# 2) Encode labels as 0/1
le = LabelEncoder()
y = le.fit_transform(melted['position'])   # x1→0, x4→1

```

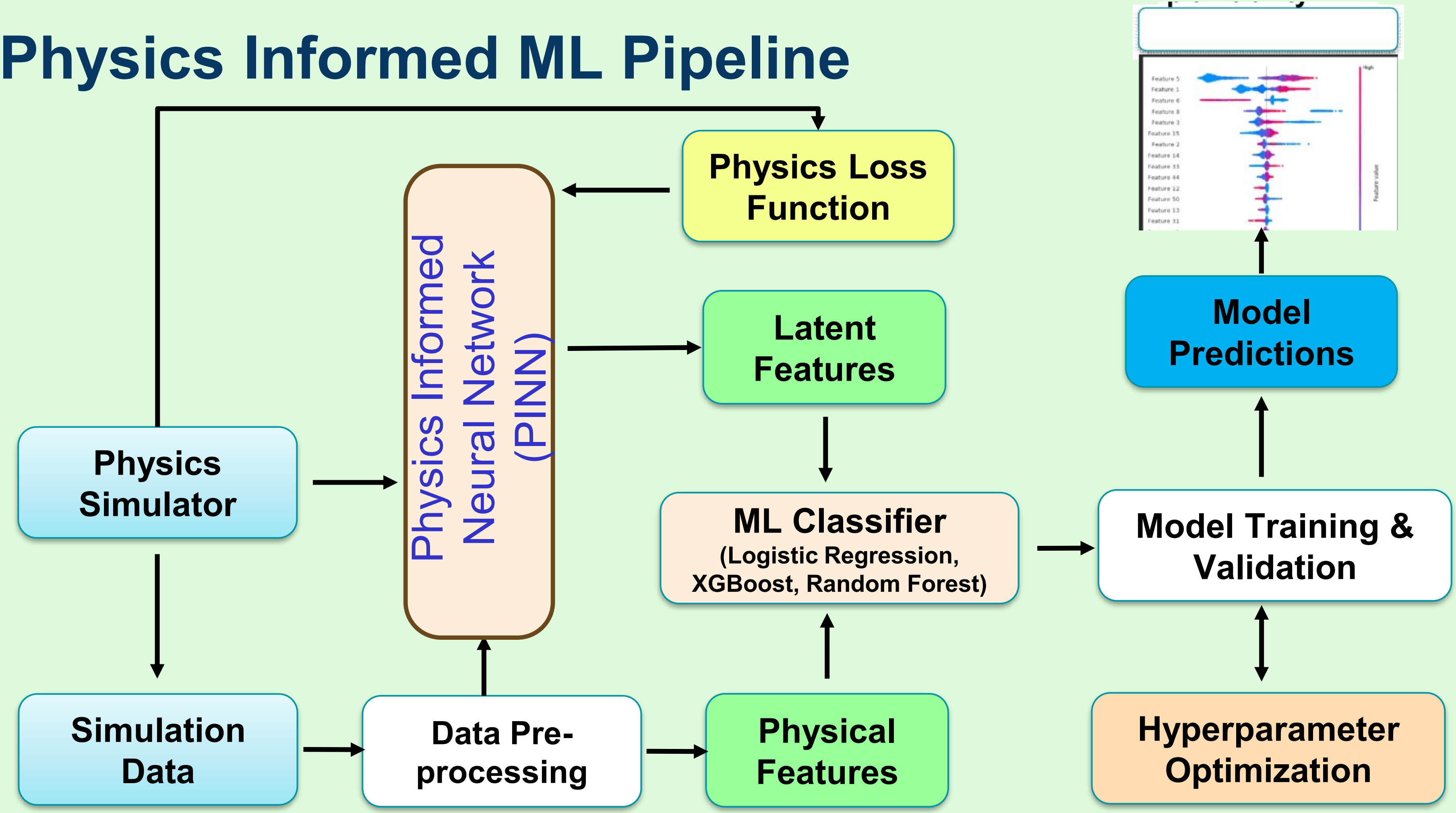
Train/Test Split

```python
# 4) Train/test split & stratify to keep classes balanced
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

Scaling

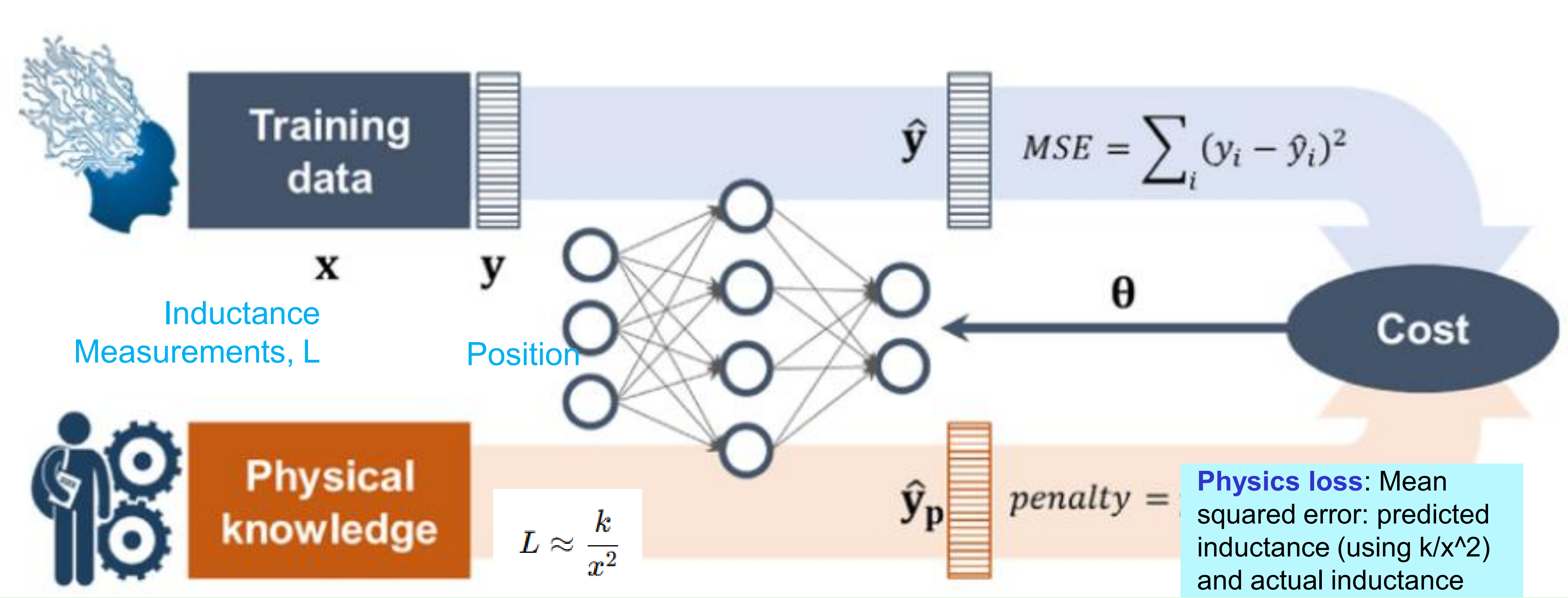
```python
scaler = StandardScaler().fit(X_train)
X_train_scaled = scaler.transform(X_train)
X_test_scaled  = scaler.transform(X_test)
```

Physics Informed ML Pipeline



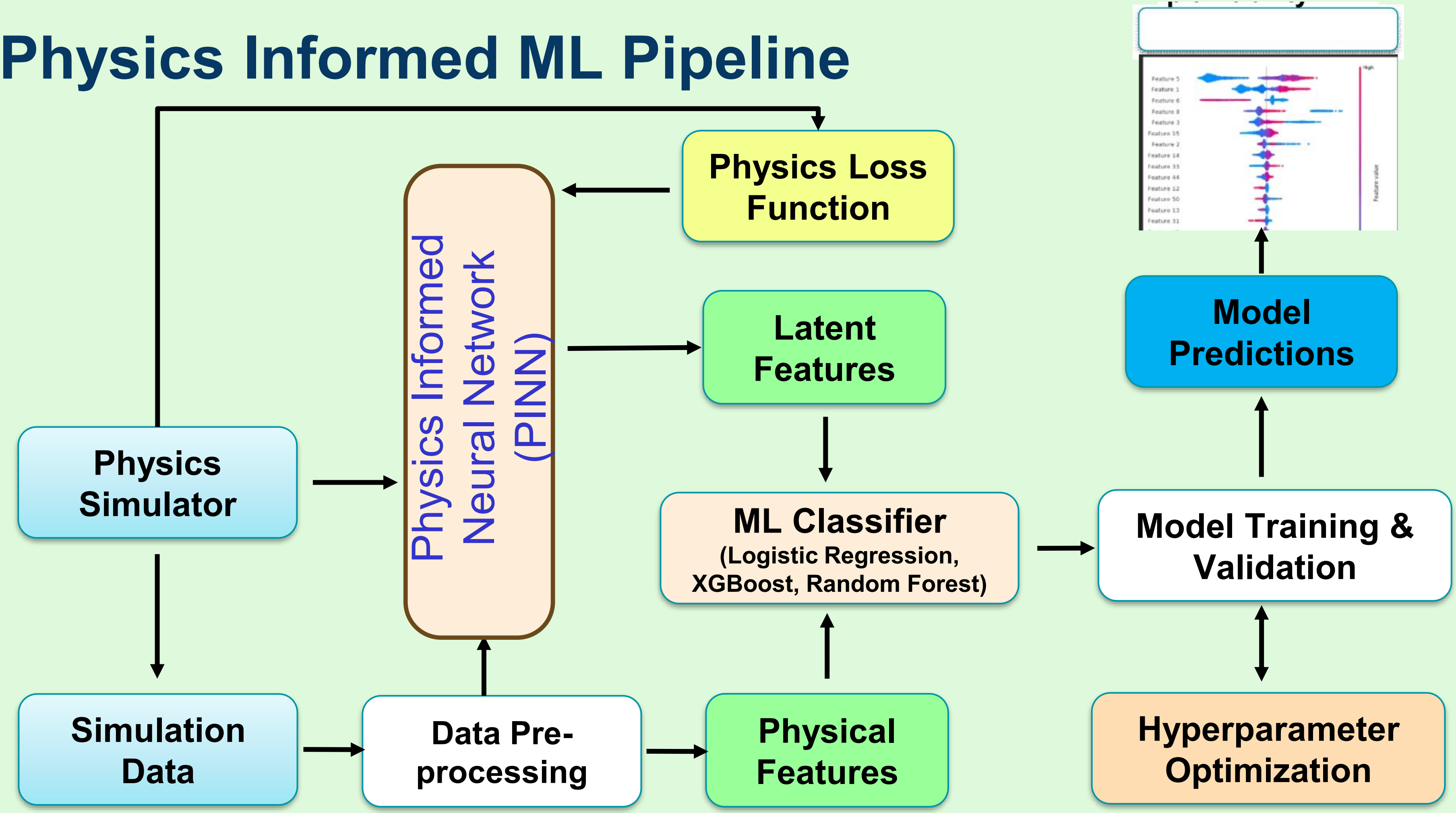
Physics Informed Neural Network (PINN)

ML Model learns a structure informed by real-world physical constraints. Latent features are more meaningful because they tie back to interpretable physics.



Source; https://web.mae.ufl.edu/nkim/Papers/paper140.pdf

Physics Informed ML Pipeline



Physical Features

1. Physics-based Position Estimate

$$x_{phys} = \sqrt{\frac{k_{opt}}{L}}$$

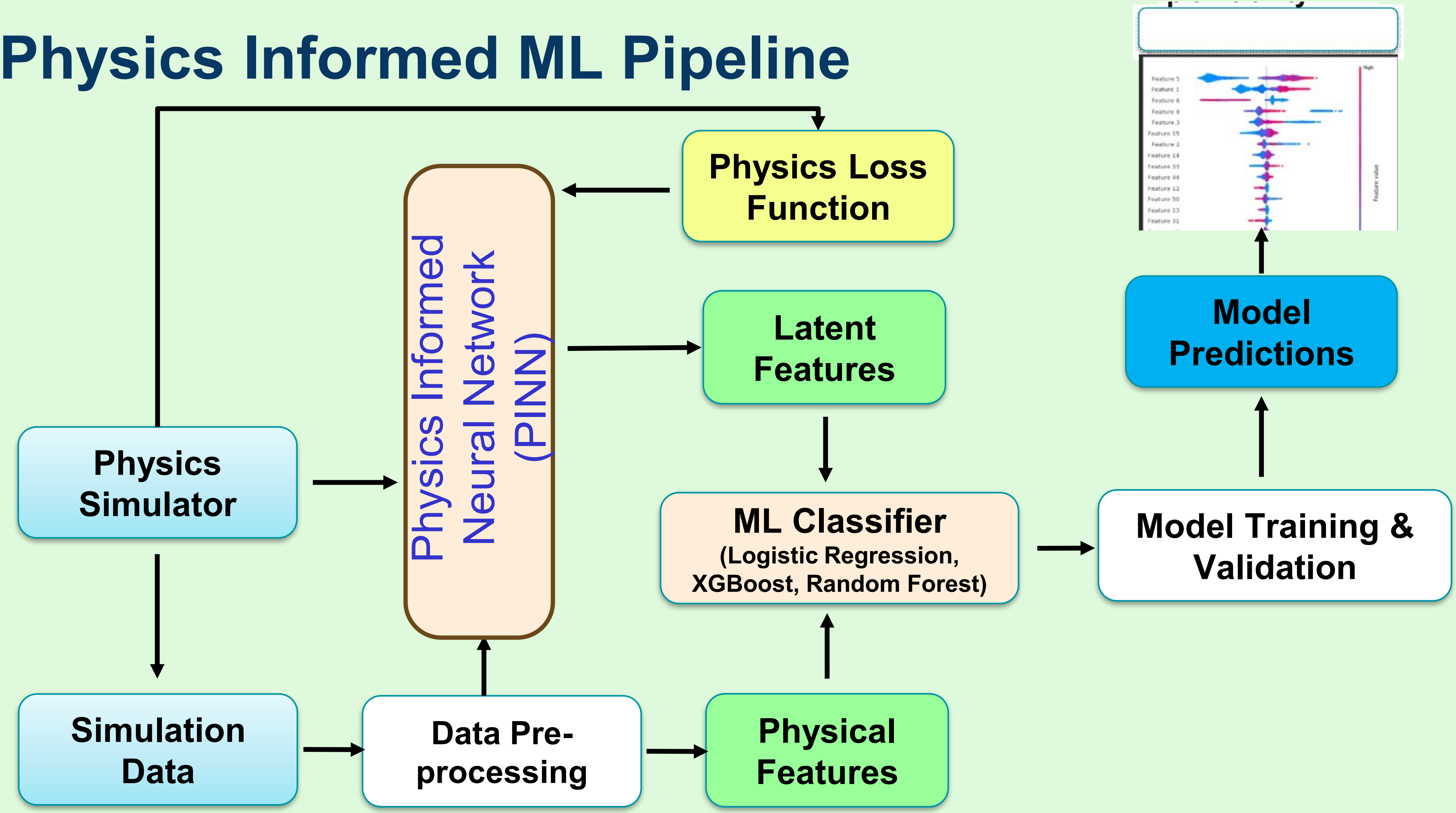
2. Relative Error between NN & Physics

$$\text{relative error} = \frac{x_{NN} - x_{phys}}{x_{phys} + \epsilon}$$

3. Implied “k” per Sample

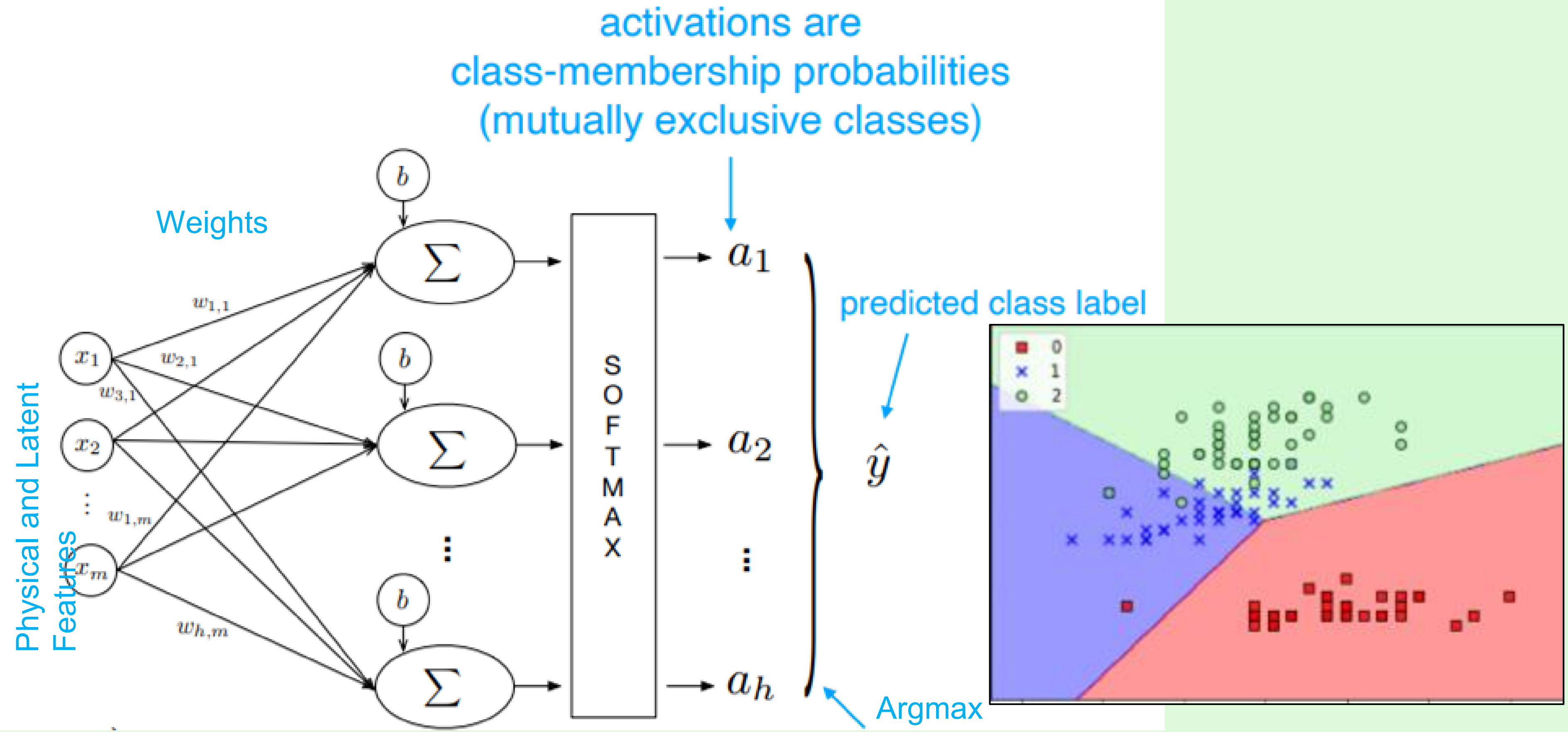
$$k_{pred} = L * x_{NN}^2$$

Physics Informed ML Pipeline

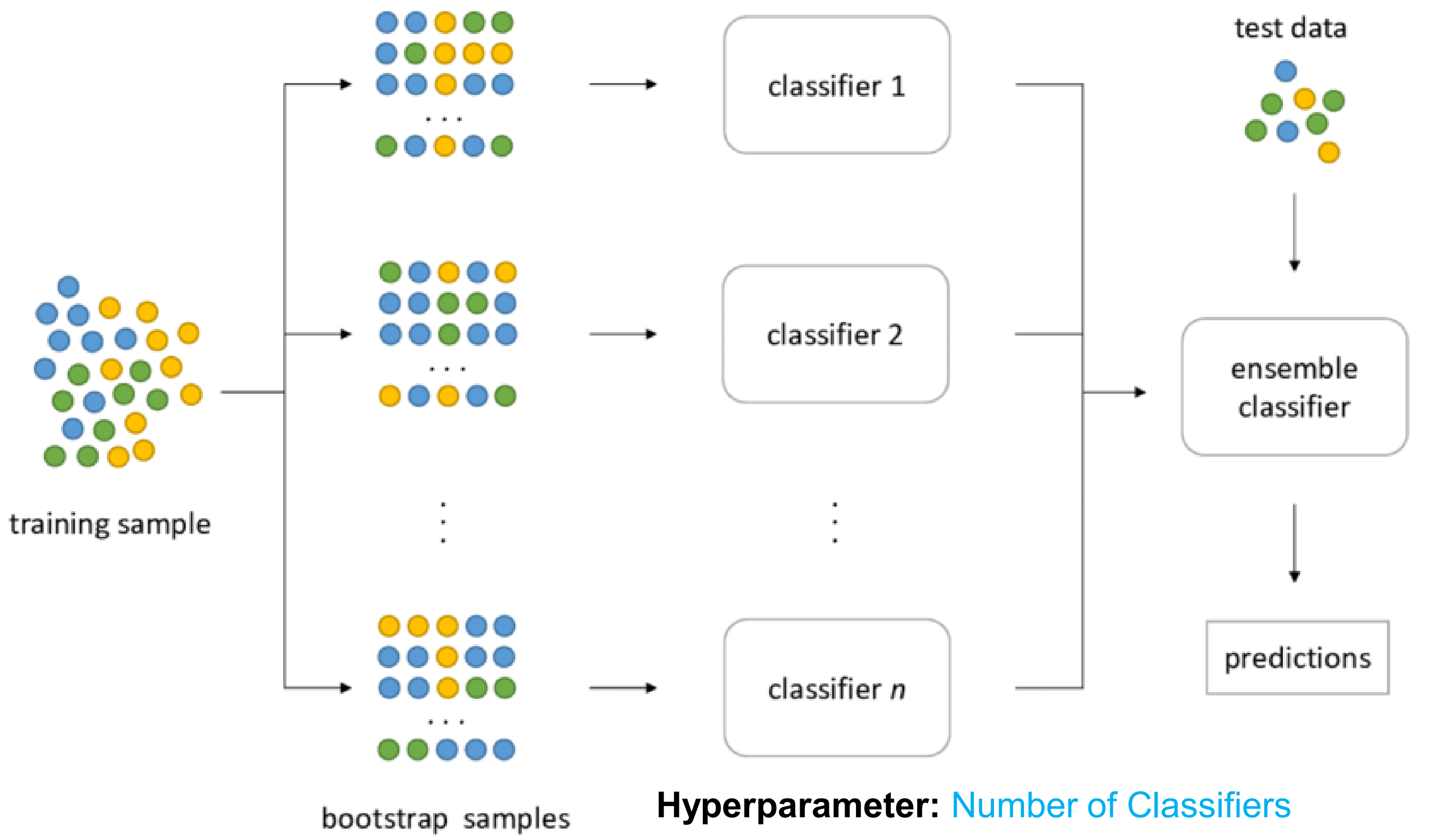


Machine Learning Models

Logistic Regression

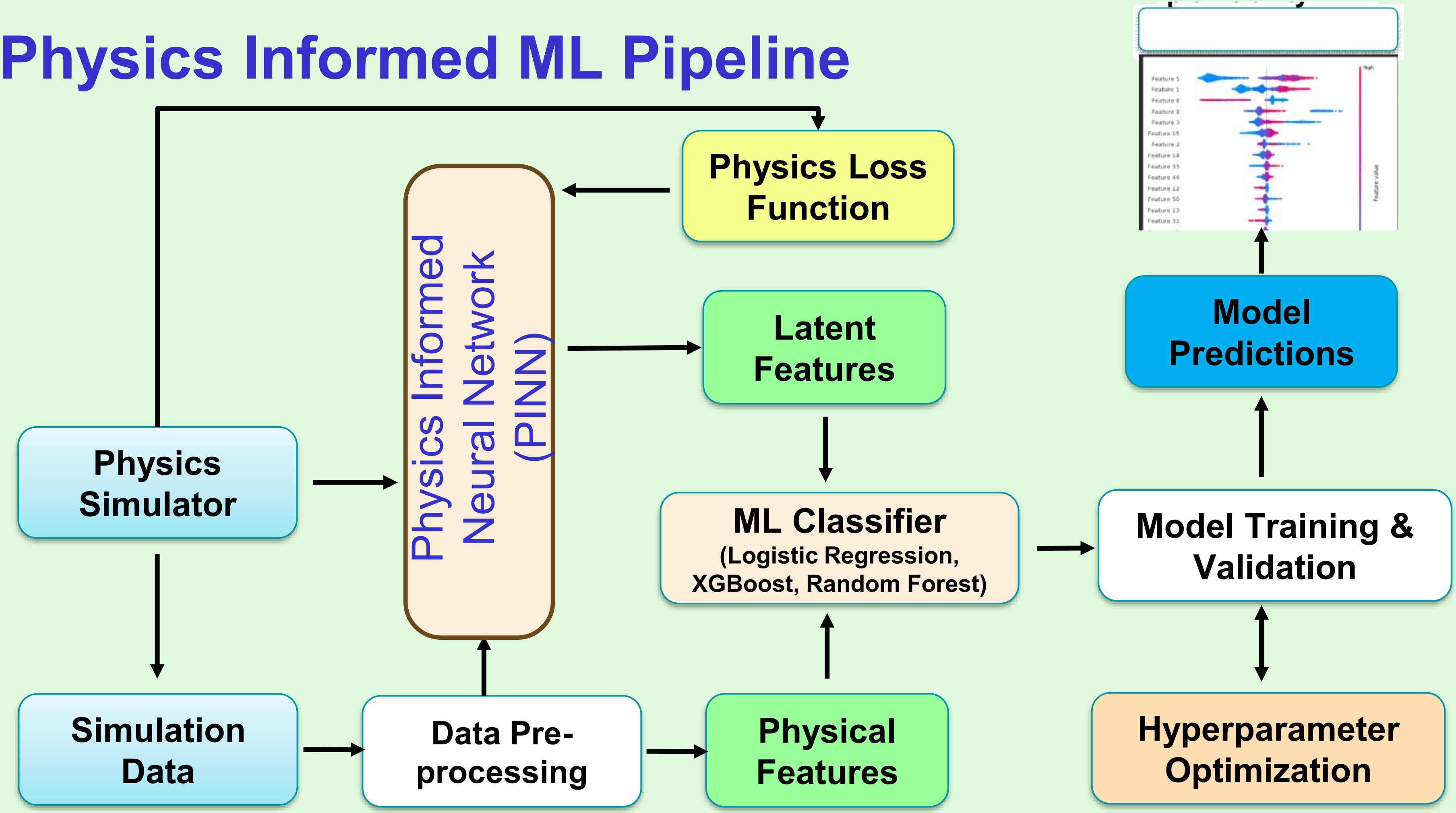


XGBoost Classifier



Hyperparameter: Number of Classifiers

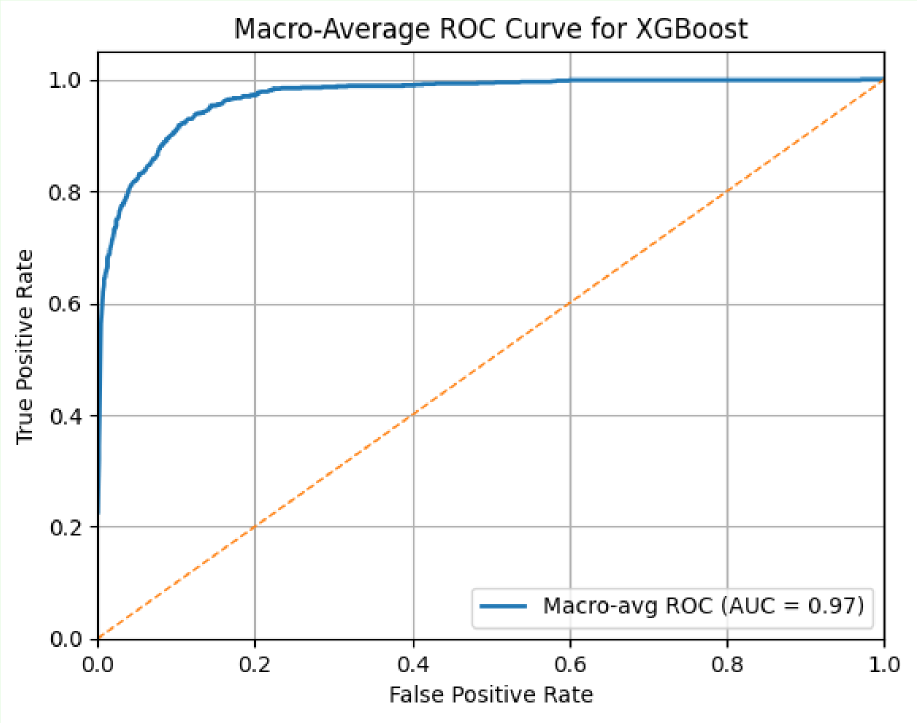
Physics Informed ML Pipeline



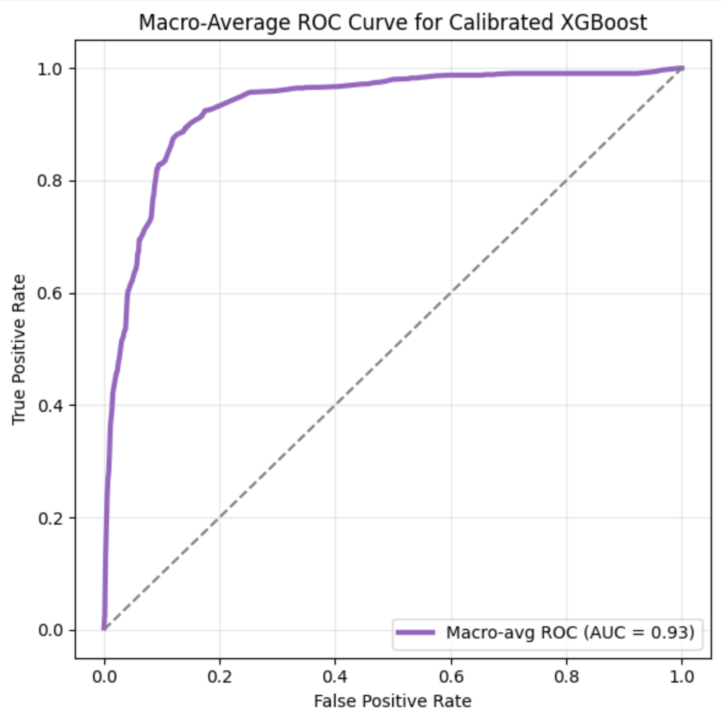
Results

XGBoost ROC Curve Comparison

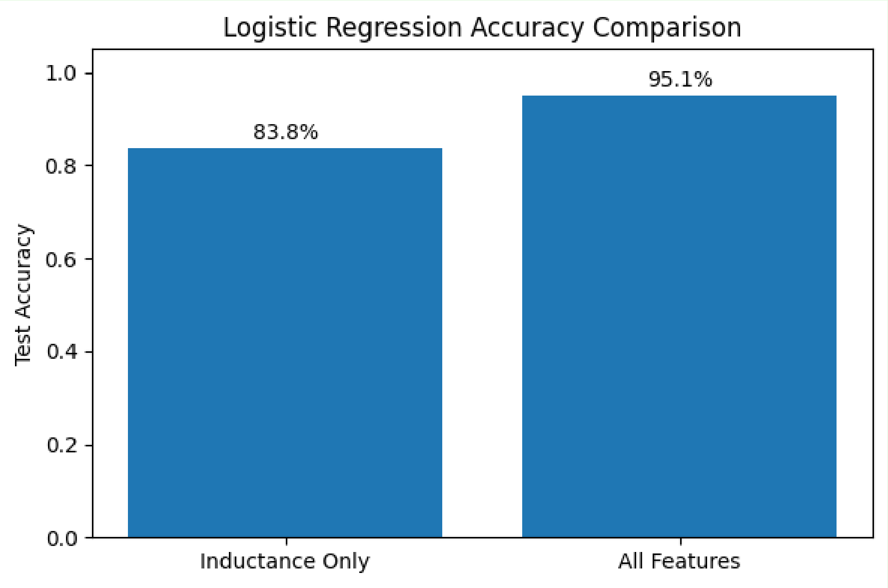
All Features



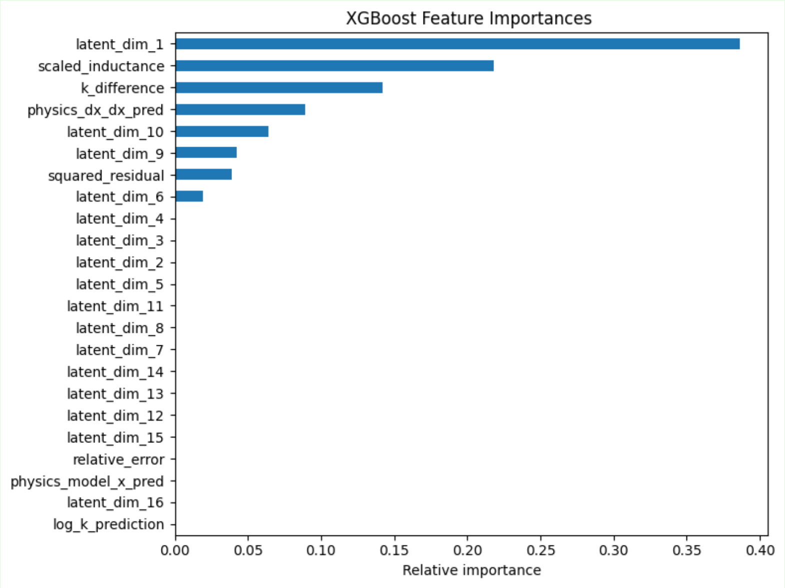
Inductance Only



Logistic Regression Accuracy Comparison



Feature Importance Map for XGBoost Model



Conclusions

- The goal of accurately detecting the position of the motorcycle was achieved
- The model with the addition features(Physical Features + PINN features) outperformed the traditional linear regression model.
- The first feature from the PINN was more important to the classification than the Induction was
- Hyperbolic Secant Noise was better at modeling data for the system than Gaussian Noise due to its tighter distribution

Future Research

- More Simulation Data
  - Is model accurate?
- Vehicle Detection
- Speed and Acceleration
- Different Geometries
  - Utilizing Optimization Techniques
  - Better Detection?
- Using Machine Learning for finding both x & y position

References

Ansys Inc. (2020). *Lesson 2: Bounding simulation space* (HFSS Getting Started 2020 R2) [PDF]. https://innovationspace.ansys.com/courses/wp-content/uploads/2021/07/HFSS_GS_2020R2_EN_LE2_BoundsSimSpace.pdf

Ansys Inc. (2020). *Lesson 3: FEM mesh and solution* (HFSS Getting Started 2020 R2) [PDF]. https://innovationspace.ansys.com/courses/wp-content/uploads/2021/07/HFSS_GS_2020R2_EN_LE3_FEM_MeshSoln.pdf

Ansys Inc. (2020). *Lesson 6: Port basics* (HFSS Getting Started 2020 R2) [PDF]. https://innovationspace.ansys.com/courses/wp-content/uploads/sites/5/2021/07/HFSS_GS_2020R2_EN_LE6_Port_Basics.pdf

Byju's. (n.d.). *Electromagnetic induction*. BYJU'S – The Learning App. Retrieved July 24, 2025, from https://byjus.com/physics/induction/

Byju's. (n.d.). *Magnetic flux*. BYJU'S – The Learning App. Retrieved July 24, 2025, from https://byjus.com/physics/magnetic-flux/

Coil32. (n.d.). *HFSS simulation of a helix coil*. Coil32 Online Calculator. Retrieved July 24, 2025, from https://coil32.net/theory/hfss-helix-coil.html

DoITPoMS. (n.d.). *Ferromagnetic materials*. Department of Materials Science and Metallurgy, University of Cambridge. Retrieved July 24, 2025, from https://www.doitpoms.ac.uk/tlplib/ferromagnetic/printall.php

Fischer, M. (2004). Skew generalized secant hyperbolic distributions: Unconditional conditional fit to asset returns. *Austrian Journal of Statistics, 33*, 183–196.

Grasp Engineering. (n.d.). *Hello world*. Retrieved July 24, 2025, from https://www.graspengineering.com/hello-world-2/

Raschka, S. (2021). *Multinomial logistic regression*. In *Intro to deep learning* [Blog post]. https://sebastianraschka.com/blog/2021/dl-course.html

Shankar, V. N. (2020). *Highway safety analytics and modeling*. Elsevier. https://www.sciencedirect.com/book/9780128168189/highway-safety-analytics-and-modeling

Siegrist, K. (n.d.). *5.31: The hyperbolic secant distribution*. In *Probability, mathematical statistics, and stochastic processes*. LibreTexts. Retrieved July 24, 2025, from https://stats.libretexts.org/Bookshelves/Probability_Theory/Probability_Mathematical_Statistics_and_Stochastic_Processes_(Siegrist)/05:_Special_Distributions/5.31:_The_Hyperbolic_Secant_Distribution

Hachcham, A. (2025). *XGBoost: Everything you need to know*. Neptune.ai. https://neptune.ai/blog/xgboost-everything-you-need-to-know

Illowsky, B., & Dean, S. (2018). *Introductory statistics*. OpenStax, Rice University.

Kerner, B. S. (2004). Three-phase traffic theory and highway capacity. *Physica A: Statistical Mechanics and Its Applications, 333*, 379–440. https://doi.org/10.1016/j.physa.2003.10.017

Keras. (n.d.). *About Keras*. Keras.io. Retrieved July 24, 2025, from https://keras.io/getting_started/about/

Kim, S., Lee, H., Choi, K., & Lee, S. (2022). Data-driven prognostics with low-fidelity physical information for digital twin: Physics-informed neural network. *Structural and Multidisciplinary Optimization, 65*(4), Article 101. https://doi.org/10.1007/s00158-022-03348-0

Klein, L. A., Mills, M. K., & Gibson, D. R. P. (2006). *Traffic detector handbook: Third edition—Volume I* (Report No. FHWA-HRT-06-108). U.S. Department of Transportation, Federal Highway Administration. https://www.fhwa.dot.gov/publications/research/operations/its/06108/02.cfm

Liao, S., Dourmashkin, P., & Belcher, J. W. (2004). *Faraday's law of induction* (Physics 8.02 TEAL Course Notes, Chapter 10) [PDF]. Massachusetts Institute of Technology. http://web.mit.edu/8.02t/www/802TEAL3D/visualizations/coursenotes/modules/guide10.pdf

LibreTexts. (n.d.). *Ferromagnetism*. Engineering LibreTexts. Retrieved July 24, 2025, from https://eng.libretexts.org/Bookshelves/Materials_Science/Supplemental_Modules_(Materials_Science)/Magnetic_Properties/Ferromagnetism

Navarro, D. (n.d.). *5.03: Skew and kurtosis*. In *Learning statistics with R: A tutorial for psychology students and other beginners*. LibreTexts. Retrieved July 24, 2025, from https://stats.libretexts.org/Bookshelves/Applied_Statistics/Learning_Statistics_with_R_-_A_tutorial_for_Psychology_Students_and_other_Beginners_(Navarro)/05%3A_Descriptive_Statistics/5.03%3A_Skew_and_Kurtosis

NVIDIA. (n.d.). *TensorFlow*. NVIDIA Glossary. Retrieved July 24, 2025, from https://www.nvidia.com/en-us/glossary/tensorflow/

OpenStax. (2016). *23.5: Faraday's law of induction—Lenz's law*. In *College Physics* (1st ed.). LibreTexts. https://phys.libretexts.org/Bookshelves/College_Physics/College_Physics_1e_(OpenStax)/23%3A_Electromagnetic_Induction_AC_Circuits_and_Electrical_Technologies/23.05%3A_Faradays_Law_of_Induction-_Lenzs_Law

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